Full-Stack Apps with Next.js

Pages, Routing, SSR, SSG, Server Actions, Caching, Tailwind, Auth, Deployment

Svetlin Nakov, PhD

Co-founder @ SoftUni

SoftUni AI https://ai.softuni.bg

Agenda

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1. **Intro to Next.js**: full-stack React framework, App Router, SSR / SSG / CSR, Turbopack
2. **Next.js Project Setup**: npx create-next-app@latest, file structure, App Router folders
3. **Pages, Routing and Layouts**: file-system routing, dynamic routes, API routes, nested layouts
4. **Client Components vs. Server Components**: "use client", hydration, SEO, data access
5. **Rendering Modes**: SSR (dynamic), SSG (static), ISR (hybrid), caching and revalidation
6. **Server Actions**: "use server", form submissions, DB mutations without REST endpoints
7. **Styling with Tailwind**: utility classes, responsive design, component styling
8. **Auth, JWT, Sessions and Middleware**: middleware.ts, roles, protected routes
9. **Performance Optimizations**: <Link> prefetch, <Image>, WebP, caching strategies
10. **Suspense and Streaming**: loading.tsx, <Suspense>, progressive HTML delivery
11. **Full-Stack App**: Neon DB + Drizzle ORM + Next.js + Tailwind, deploy to Netlify

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Breaks

20:00 / 21:00

Introduction to Next.js

Full-Stack React Framework for Modern Web Apps, with Server-Side Rendering

What is Next.js?

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- **Next.js** is a **full-stack framework** for modern Web apps
  - Built on top of **React**, by **Vercel** – https://nextjs.org
  - Adds **routing**, **server rendering**, **data fetching**, **caching**

- Why Next.js apps are **fast**?
  - **Server rendering** → server sends HTML → faster load + SEO
- **Front-end** + **back-end** into a single app (React + TS)
  - **Front-end**: React components + UI rendering
  - **Back-end**: server actions + API routes
  - Direct **database access** from server components

Why Next.js for React Devs?

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⚛ Plain React Apps

Client-side rendering

- **SEO-unfriendly** – search engines initially see empty HTML
- **Slower first paint** – JavaScript must download and execute first
- **Manual setup** – routing, data fetching, builds require configs
- **No server-side capabilities** – no built-in backend or SSR support
  - Apps need separate back-end

▲ Next.js Apps

Server rendering + full-stack

- **SEO-friendly** – content comes as HTML from the server
- **Faster initial load** – pre-rendered pages improve performance
- **Built-in routing** – no manual setup
- **Built-in optimizations** – image handling, caching, and code splitting
- **Serverless deployment** – back-end + front-end both run in Vercel / Netlify

Rendering Modes in Next.js

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🌐 CSR (Client-Side Rendering)

- Browser receives **empty HTML + JS**, then **React builds the UI**
- **Not SEO-friendly**, slower first paint

⚙️ SSR (Server-Side Rendering)

- Server builds **HTML on each request**, sent ready to browser
- **SEO-friendly**, always fresh data

📄 SSG (Static Site Generation)

- HTML built **once at build time**, served from CDN
- **Fastest**, SEO-friendly, ideal for marketing pages

🔄 ISR (Hybrid Rendering)

- **Incremental Static Regeneration** — best of SSG + SSR
- **Static**, but auto-**revalidates** in the background (e.g. in 5 mins)

Next.js Key Components

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- **App Router** – file-system based routing in the **app/** folder
- **React** – latest React with **server components** support
- **Turbopack** – next-gen Rust-based bundler, fast and stable
- **Server Components** – render on the server, can access DB directly
- **Client Components** – render in the browser, handle interactivity
- **API Routes** – backend endpoints inside Next.js
- **Server Actions** – call server functions directly from forms
- **Middleware** – runs before each request (auth, redirects, logging)
- **Built-in optimizations** – **<Image>**, **<Link>**, fonts, caching
- **Streaming & Suspense** – progressive UI loading (with indicator)

Next.js Versions

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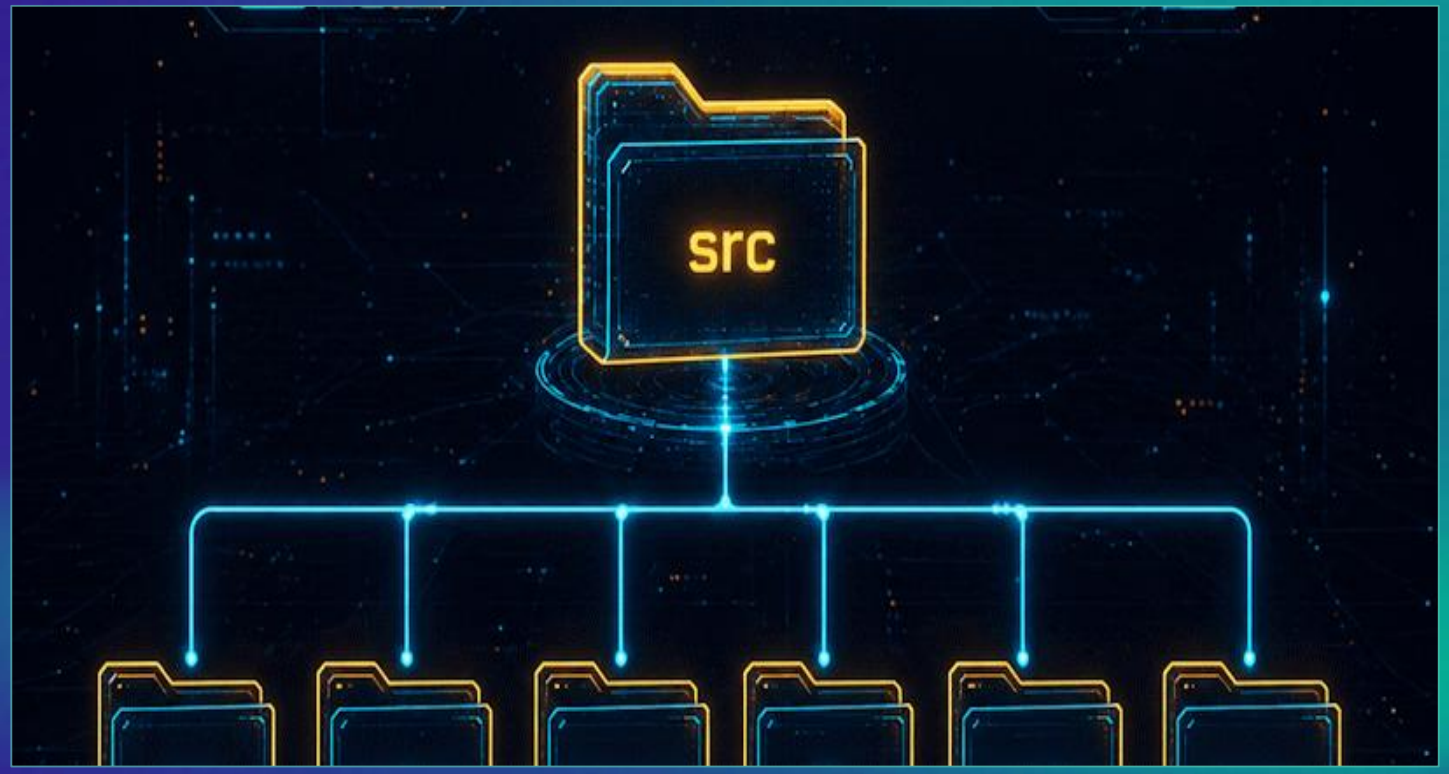
- Next.js has two major variants:
  - **Pages Router** (Next.js < v13) | **App Router** (Next.js v13+)
- **Old Next.js**: pages router with client-first rendering in **/pages**, no server actions → legacy, don't use for new projects
- **New Next.js**: app router with server-first rendering in **/app**, nested layouts, server actions, API routes, loading / error UI
  - Use the **modern Next.js**, with server-side rendering!
- Note: **AI dev agents** can **confuse** Next.js versions → incompatible code / config / libraries, hallucinations

Next.js Project Setup

Scaffolding, File Structure, App Router



Creating a Next.js Project

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🚀 Scaffold a **Next.js** project with the official CLI:

```
npx create-next-app@latest .
```

▶️ Then **run the dev server** → http://localhost:3000

```
npm run dev
```

⚙️ Recommended setup:

- **TypeScript**: Yes ✓
- **ESLint**: Yes ✓
- **Tailwind**: Yes ✓
- **App Router**: Yes ✓
- **Turbopack**: Yes ✓

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS nextjs-examples> npm run dev
> next dev

▲ Next.js 16.2.4 (Turbopack)
- Local:         http://localhost:3000
- Network:       http://10.5.0.2:3000
✓ Ready in 388ms
```

Running a Next.js Project

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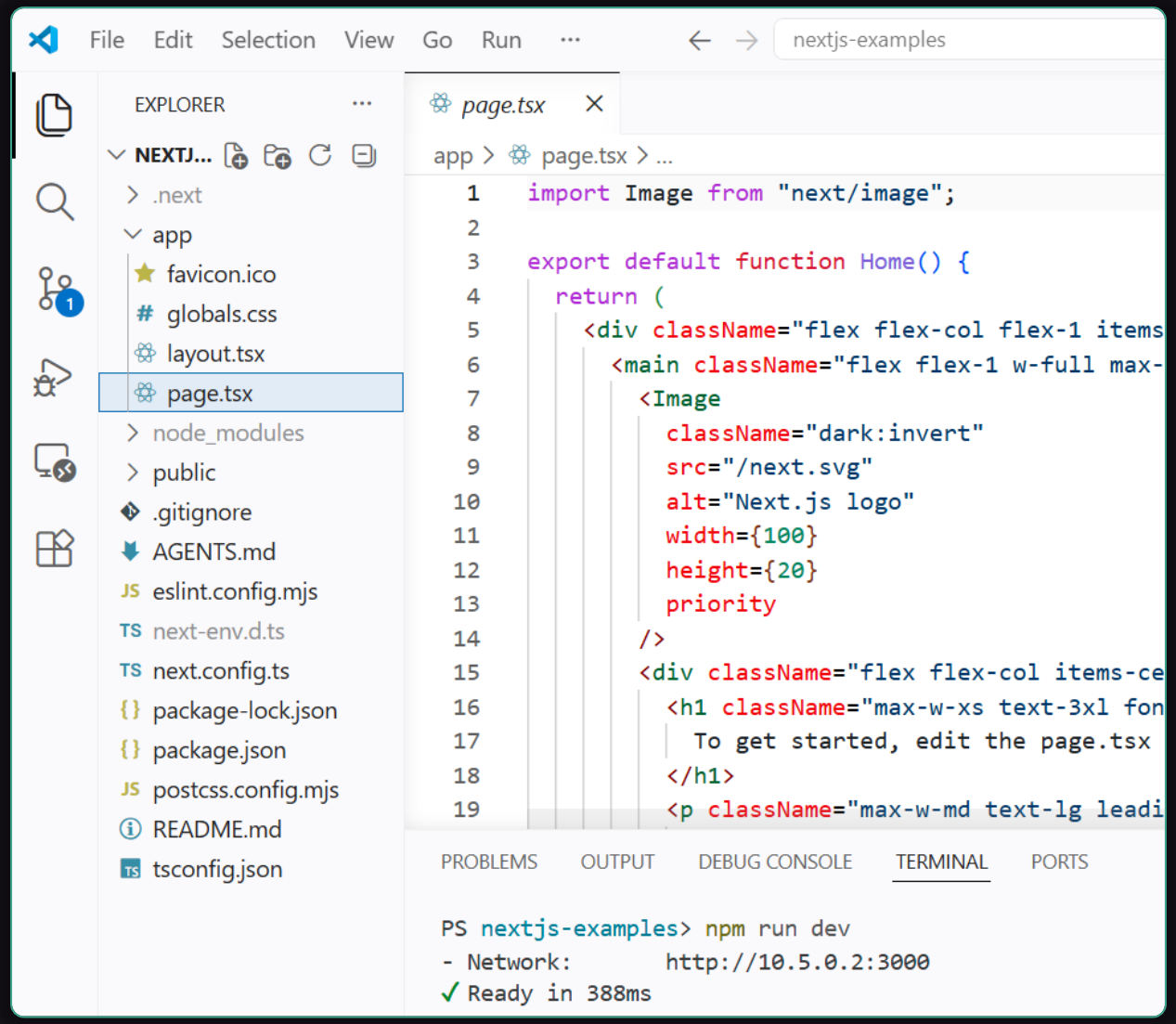

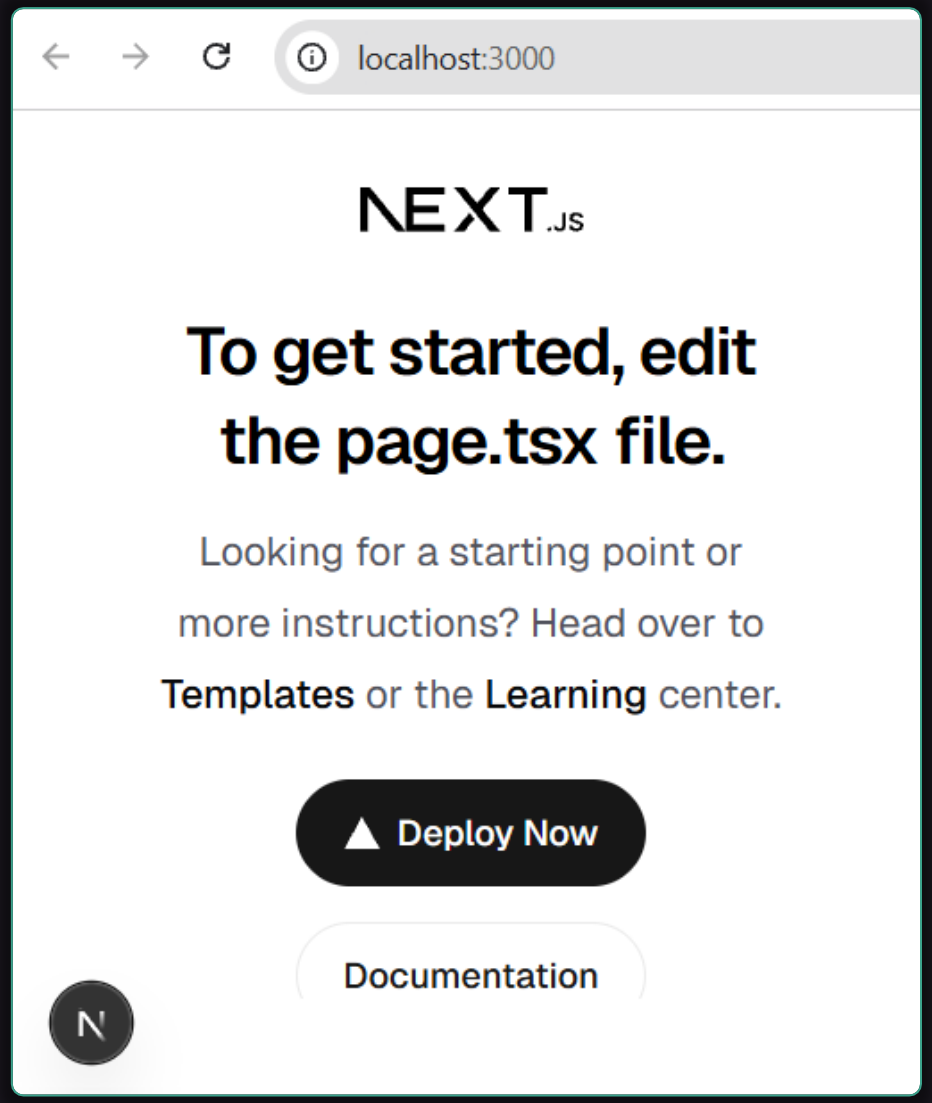

Next.js – Project Structure

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- ⚙️ `package.json` – Node.js config
- 📁 `app/` – routes, layouts, pages
- 📄 `layout.tsx` – root layout
- 📄 `page.tsx` – home page "/"
- 🖼️ `public/` – static assets
- 🧩 `components/` – reusable React components
- 🛠️ `lib/` – helpers, utils, db client
- 🔐 `.env` – secrets (DB URL, JWT)
- ⚙️ `next.config.ts` – Next.js config
- ⚙️ `tsconfig.json` – TypeScript config

```
my-nextjs-app/
├── package.json
├── app/
│   ├── layout.tsx
│   ├── page.tsx
│   └── globals.css
├── public/
├── components/
├── lib/
├── .env
├── next.config.ts
├── tsconfig.json
└── AGENTS.md / CLAUDE.md
```

Live Demo – First Next.js App

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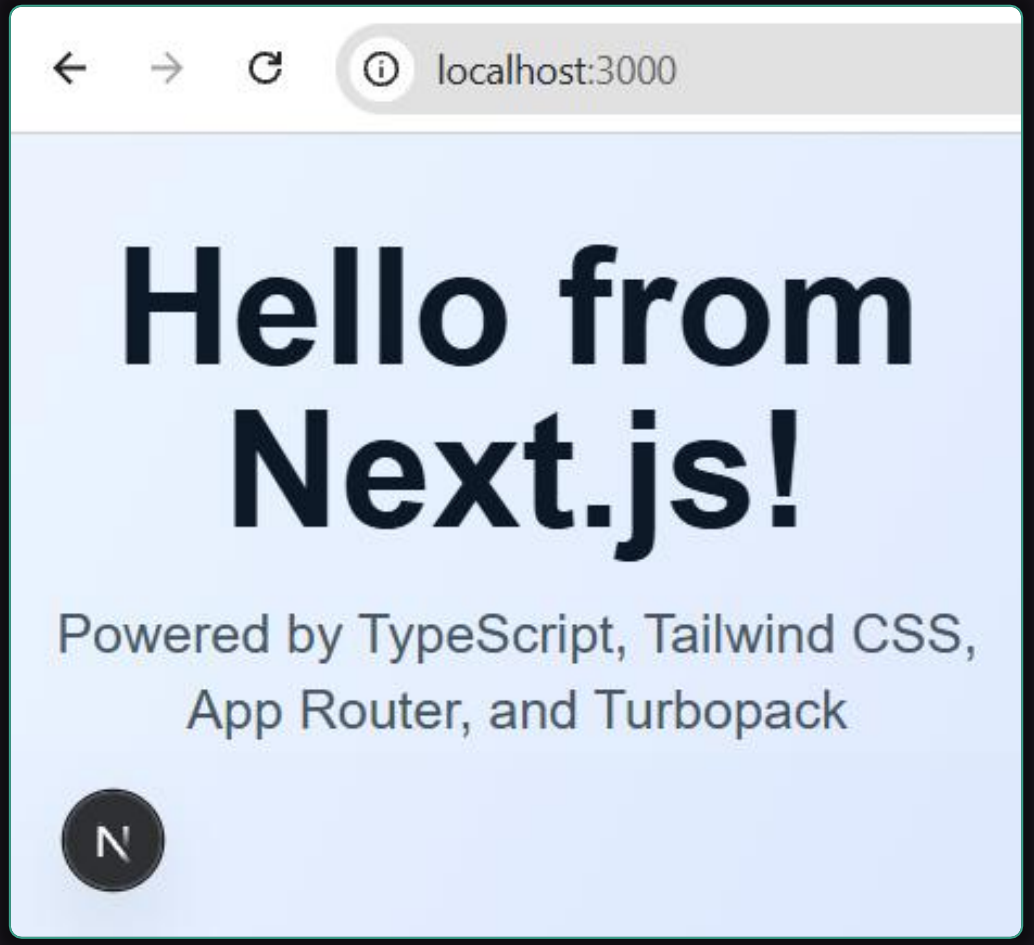
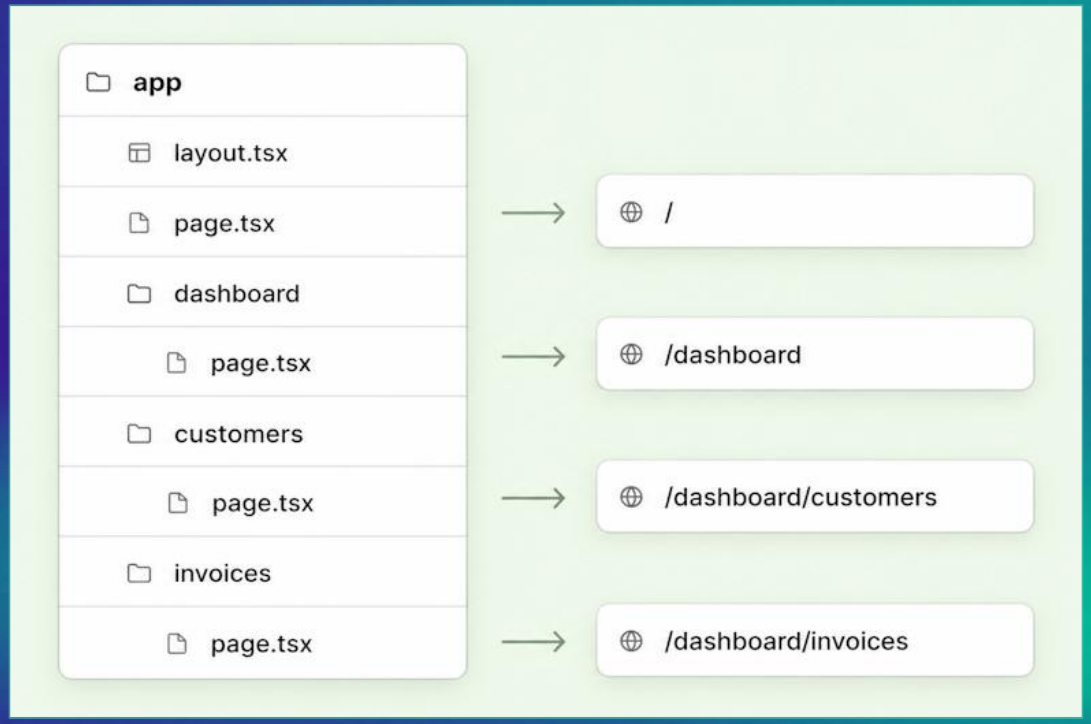
🤖 **AI prompt** to scaffold and run a starter Next.js app:

Scaffold a **new Next.js app**:
- Use **TypeScript**, **Tailwind**, **App Router**, **Turbopack**
- Edit `app/page.tsx` to show: "*Hello from Next.js!*"
- Start the **dev server** and open the app in the browser



Pages, Routing and Layouts

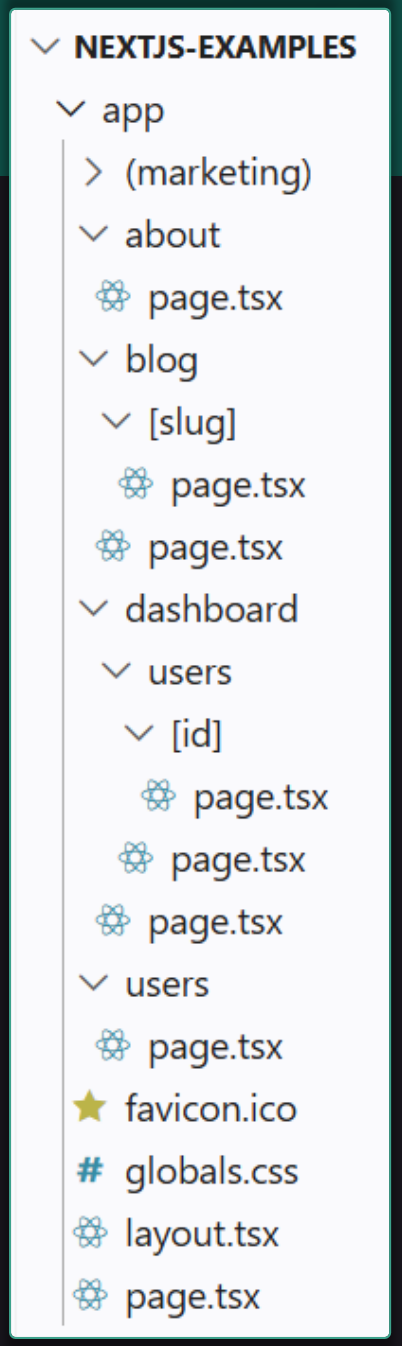
App Router, File-System Routing, Dynamic Routes, API Routes



File-System Routing

- In Next.js each **folder** in `app/` = a route segment
  - `page.tsx` = the UI shown at that URL
  - Brackets `[id]` = **dynamic segment**
- **No router config** — the file tree is the router

```
app/page.tsx                          →  /
app/about/page.tsx                    →  /about
app/blog/page.tsx                     →  /blog
app/blog/[slug]/page.tsx              →  /blog/hello
app/dashboard/page.tsx                →  /dashboard
app/dashboard/users/page.tsx          →  /dashboard/users
app/users/[id]/page.tsx               →  /users/42
app/(marketing)/page.tsx              →  (group, no URL)
```



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Creating a Simple Page

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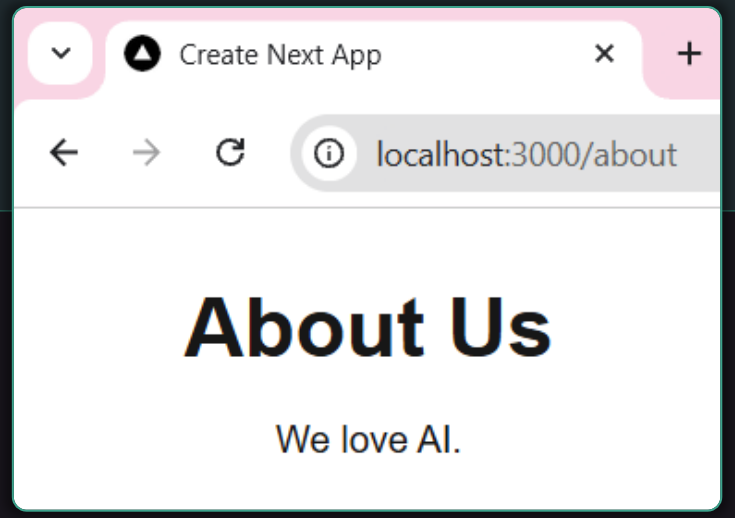
🤖 **AI prompt** to create a new app page:

Create a **new page**: **/about**
- Heading **<h1>**: About Us
- A paragraph: about text
- Style with **Tailwind**: centered, padded

```
export default function AboutPage() {
  return (
    <main className="p-8 text-center">
      <h1 className="text-4xl font-bold">
        About Us
      </h1>
      <p className="mt-4">We love AI.</p>
    </main>
  );
}
```



Dynamic Routes

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🛣️ Folders wrapped in `[brackets]` become **route parameters**

- 📂 `app/news/[id]/page.tsx` matches **/news/1**, **/news/42**, …
- ⚡ `params` is **async**

🌐 Use `[...slug]` for **catch-all routes**

- **app/docs/[...slug]/page.tsx** matches **/docs/react/hooks**, **/docs/react/hooks/use-effect**

```
type Props = {
  params: Promise<{ id: string }>;
};
```

```
export default async function
NewsItem({ params }: Props) {
  const { id } = await params;
  return <h1>Article #{id}</h1>;
}
```

Live Demo – Page with Route

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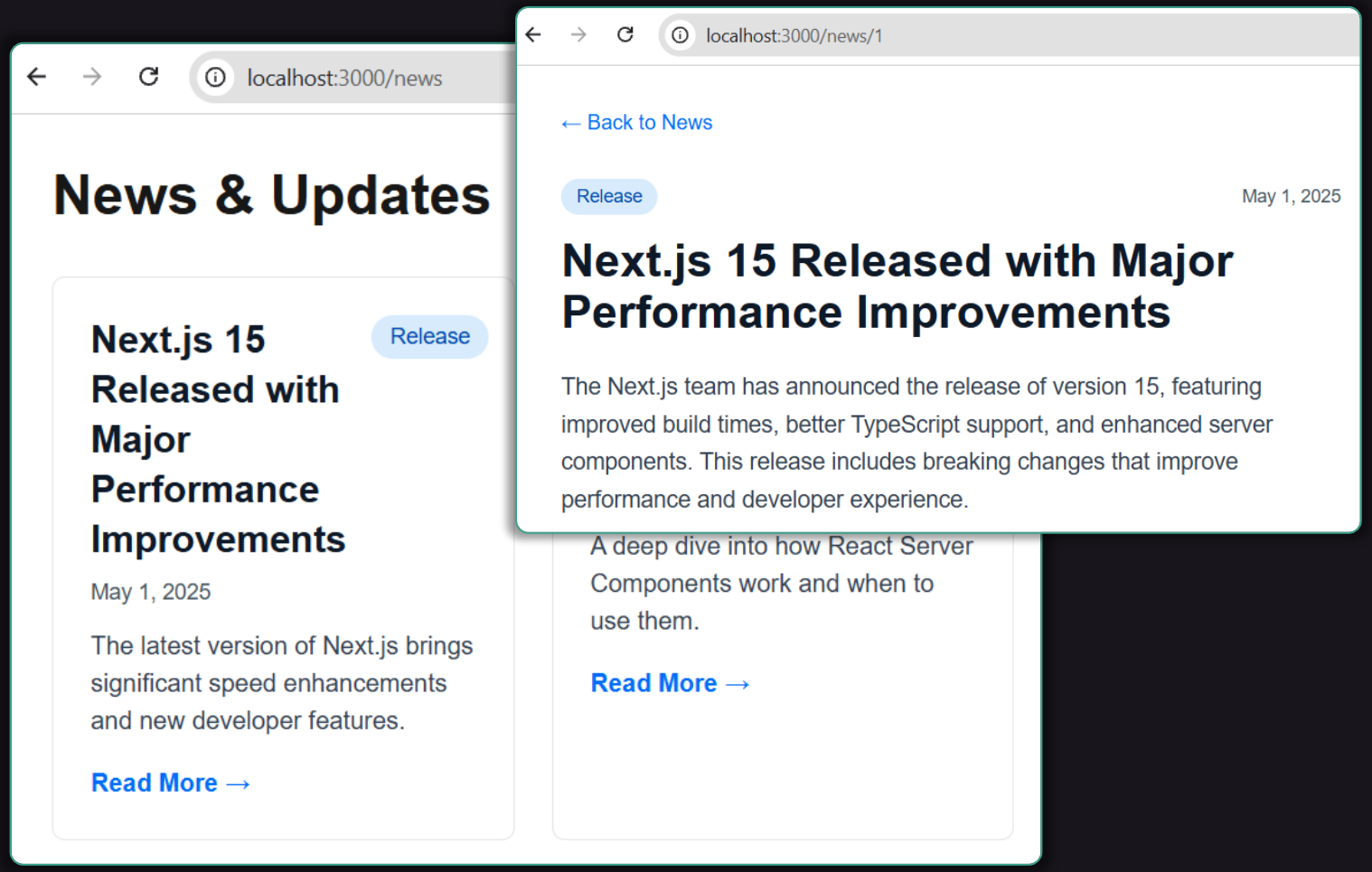
🤖 **AI prompt** to create a page with dynamic routing:

Create a page **/news** → render a list of **news**

Create a dynamic route: **/news/[id]** → render **news item**

Define sample list of news in `**news-data.ts**`.



API Routes

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- Server-only endpoints in `app/api/…/route.ts`
- File exports HTTP method functions: `GET`, `POST`, `PUT`, `DELETE`
- Call from the client with `fetch("/api/hello")`

```
export async function GET() {
  return NextResponse.json({
    message: "Hello API!"
  });
}
```

```
export async function POST(req: Request) {
  const body = await req.json();
  return NextResponse.json({body });
}
```

```
async function loadApiHello() {
  let data = await (await fetch("/api/hello")).json();
  setMessage(data.message);
}
```

Live Demo – API Routes

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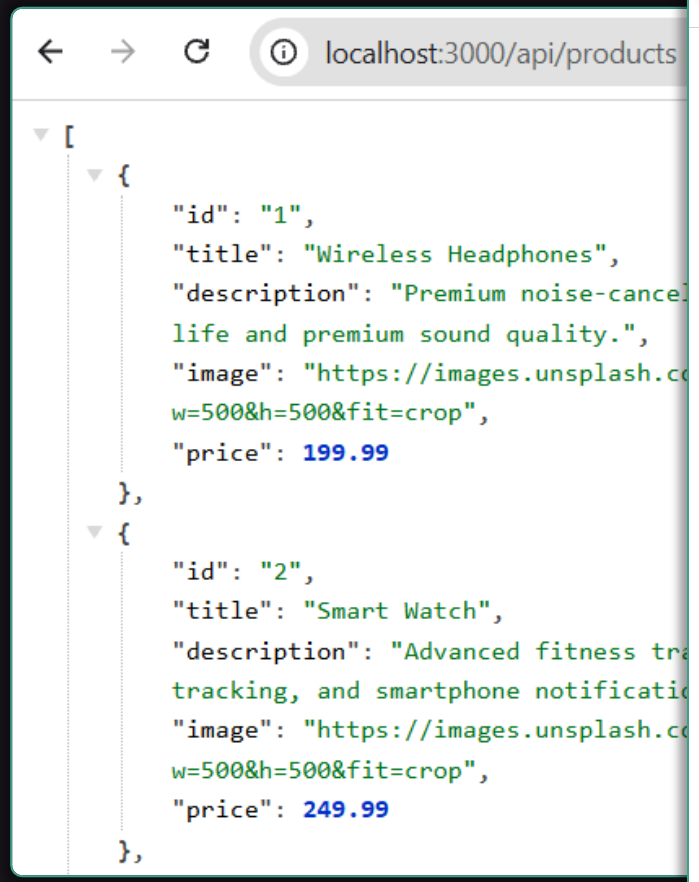
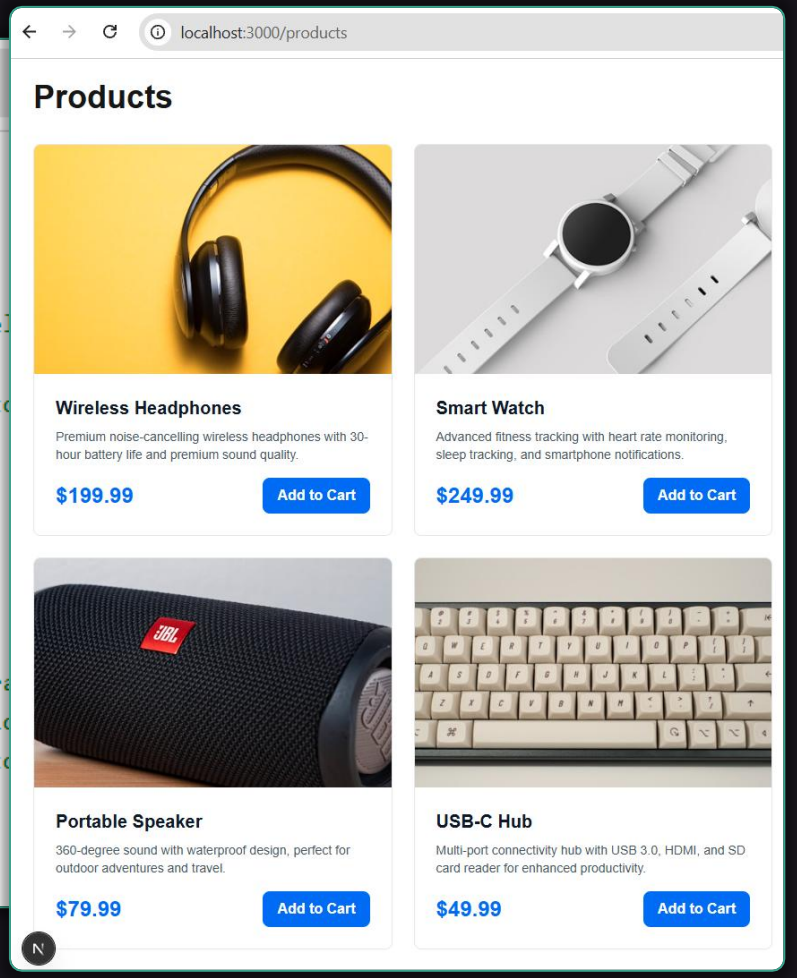
🤖 AI prompt to create API routes + client page:

Create an API route **GET /api/products** → return a list of products

Define sample products in `**products-data.ts**` (title, description, image)

Create a page **/products** → display the products from the API as cards





Layouts

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- `layout.tsx` = shared UI wrapper for a route + children
  - **Root layout** `app/layout.tsx` wraps the entire app
  - **Nested layouts**: `app/dashboard/layout.tsx` wraps `/dashboard/*`
- **Layouts persist** on navigation — no re-render of the wrapper

```
export default function
RootLayout({ children }: {
  children: React.ReactNode;
}) {
  return (
    <html lang="en">
      <body>
        <Header />
        {children}
        <Footer />
      </body>
    </html>
  );
}
```

Live Demo – Site Layout

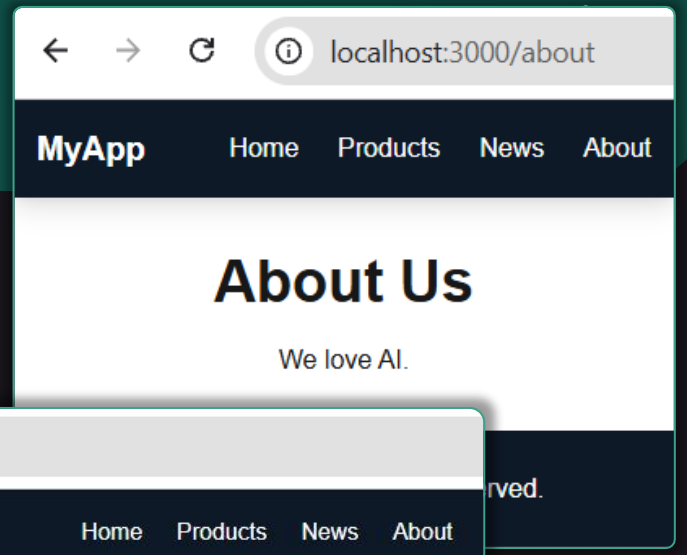
🤖 **AI prompt** to create site layouts:

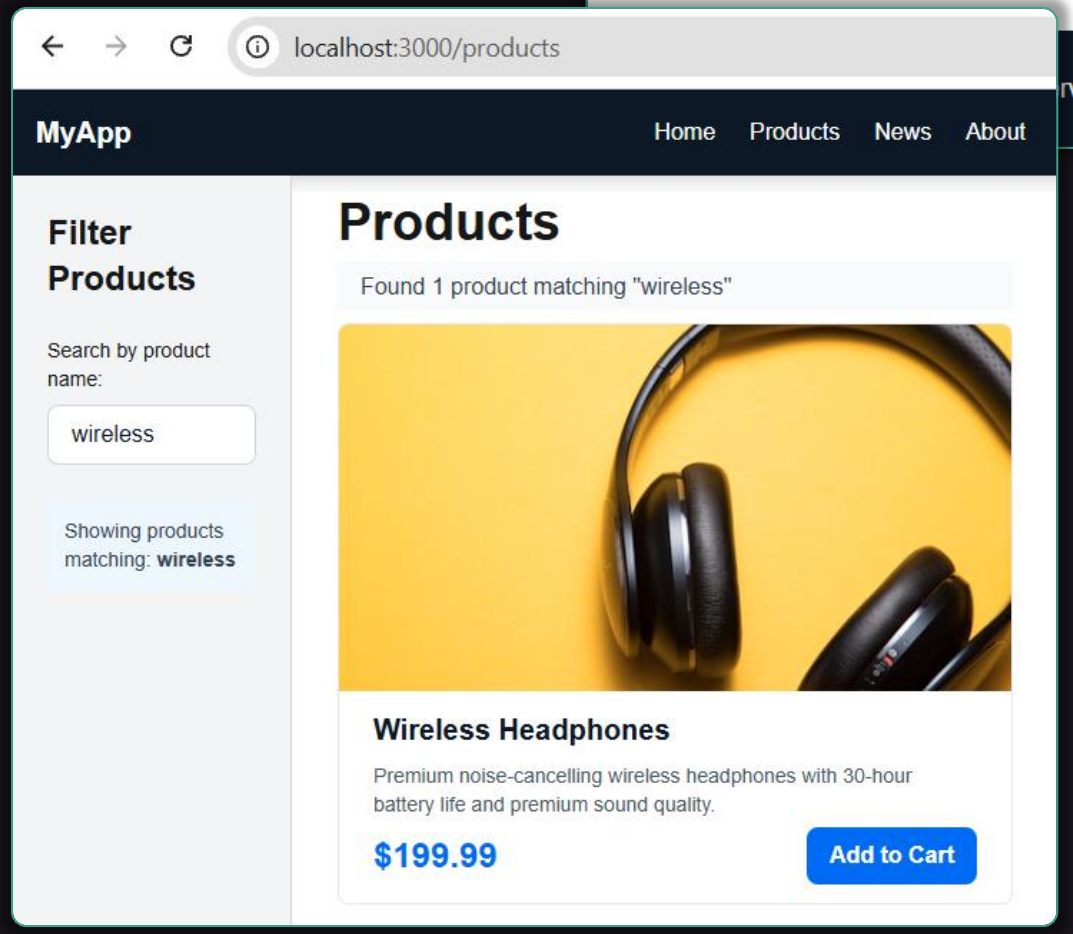
Implement a **site layout**:

- Top **nav bar with links**: Home, Products, News, About
- **Footer** with copyright notice
- Apply to all pages via `app/layout.tsx`

Create **nested layout** at `app/products/layout.tsx`

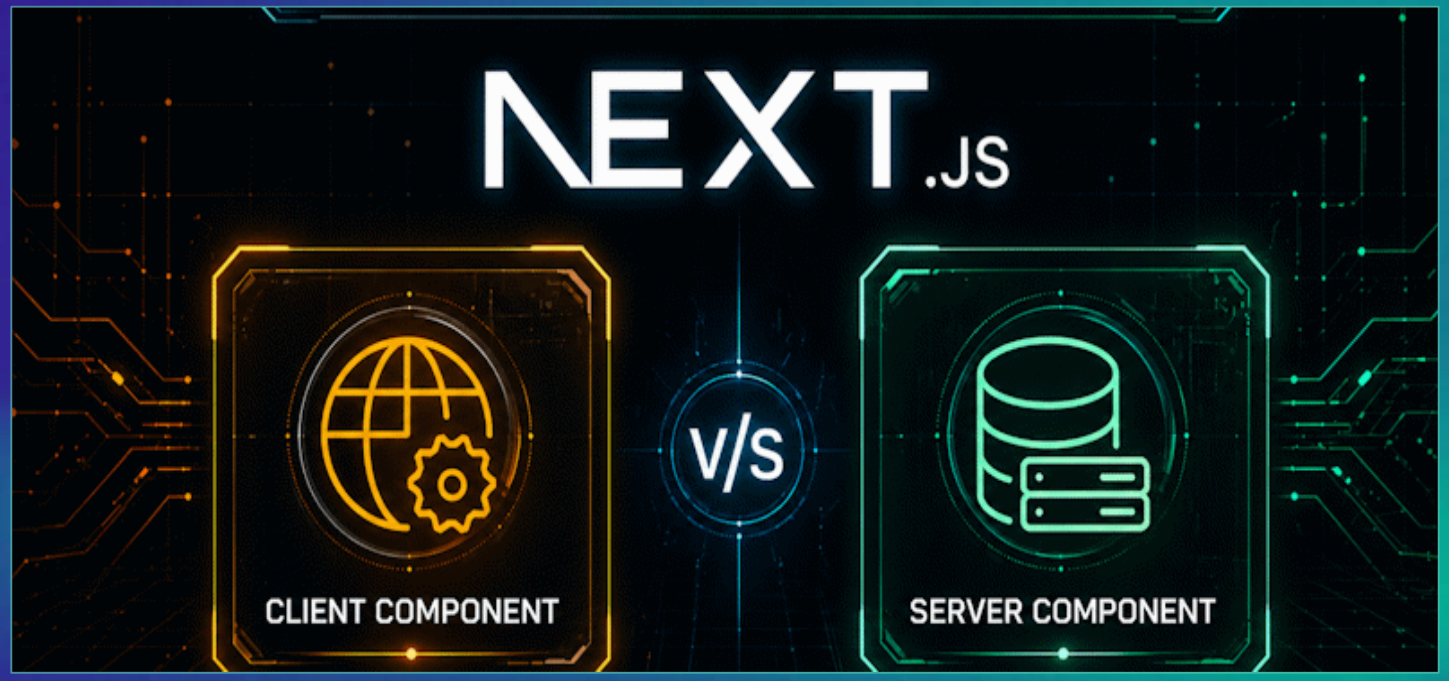
- Display a sidebar: filter products by substring





Client Components vs. Server Components

Rendering Modes, Hydration, SEO, Data Access



Server and Client Components

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- **Client components**
  - Run in the **browser**
  - Server sends **HTML+JS code**
  - **React** renders / updates the UI in the Web browser
- **Server components** (default in Next.js)
  - Run on the **server**
  - Server sends **HTML code** + component tree (no JS)
  - **Browser** renders the HTML

```
"use client"
export function Button() {
  return <button onClick={() =>
   alert("Hi")}>Click</button>
}
```

```
export default async function Page() {
  const posts = await getPosts();
  return <div>{posts.count} posts</div>
}
```

```
export async function getPost() {
  return db.posts.select();
}
```

Server Components by Default

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📌 **By default**, components in Next.js are "**server components**"

- Unless declared **"use client"**, React components run server-side

🖥️ **Server components**:

- Render on the server → **HTML sent to the browser**
- **SEO-friendly** – content visible in **[View Page Source]** / **[Ctrl+U]**
- Can access the **database** and **app secrets** directly
- **Automatically cached** by default (unless "no-cache")
- Cannot use `useState()`, `useEffect()`, or browser APIs
- Cannot handle browser events like `` `onClick` ``

Server Component – Example

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- **Server components** run on the server
  - **Query DB directly** → return HTML
  - No `useEffect()`, no `fetch('/api/…')` needed
  - The component is `async` → await DB queries inline

```
// app/books/page.tsx (Server Component)
import { db } from "@/lib/db";
```

```
export default async function BooksPage() {
  const books = await db.select()
    .from(booksTable);
  return (
    <ul>
      {books.map((b) => (
        <li key={b.id}>{b.title}</li>
      ))}
    </ul>
  );
}
```

Client Components

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- **Client components**
  - Declared with `"use client"` at the top
  - Ship as HTML + JS to the browser → **hydrated** by React to add interactivity
  - Used for **client-side interactions**
  - Can use `useState()`, `useEffect()`, `onClick`, browser APIs
  - **NOT SEO-friendly** — interactive parts render in the browser
  - Use for: **forms**, **buttons**, **modals**, **charts** — anything interactive
- **Rule**: keep the client bundle small
  - Use client components **only where needed** (client-side interactions)

```
"use client";
export default function ClickMe() {
  return (
    <p onClick={() =>
      console.log("Msg in the
      browser")}>Click me</p>
  );
}
```

Client Component – Example

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- Marked with `"use client"` → runs in the **browser**
- A server page can still **import and render** `<Counter />`

```
// app/page.tsx
<h1>Home Page</h1>
<Counter />
```

- Server sends initial HTML + JS bundle → React **hydrates** in browser

```
// app/components/Counter.tsx
"use client";
import { useState } from "react";

export default function Counter() {
  const [count, setCount] =
    useState(0);
  return (
    <button onClick={() =>
        setCount(count + 1)}>
      Clicked {count} times
    </button>
  );
}
```

Server vs. Client – Side by Side

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🖥️ Server Component

used by default in Next.js

- Rendered on the **server**
- `async function` allowed at top level
- No hooks, no events
- **Direct DB** / **API** / **files** access
- **SEO-friendly** HTML
- Better **performance** (load faster)

💻 Client Component

"use client"

- Pre-rendered at the **server**
  - **Hydrated** in the browser
- No `async` at top level
- Uses hooks: `useState()`, `useEffect()`
- Handles browser events: `onClick`, `onChange`

Mix freely: server components can import client components (not vice-versa)

Hydration Explained

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- **Hydration** = making static HTML interactive
- **Client components flow**:
  1. 🖥️ **Server renders HTML** → sent to browser
  2. ⚡ Browser displays static HTML — **fast first paint**
  3. 📦 **JS bundle downloads** (component JS code)
  4. ⚛️ React **hydrates** → attaches event listeners + restores state
  5. ✨ Page becomes **interactive**
- Only **client components** need hydration
- **Server components = pure HTML**, zero JS shipped for them

Mixing Server and Client Components

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- When to use **server components**?
  - Page main content
  - Texts, visible for search engines: blog posts, news, product descriptions
  - Layouts, headers, footers, sidebars
  - All other page content
- When to use **client components**?
  - Only when server-rendering does not work
  - Interactive elements: client-side filters, modal popups, tabs, like box, …
  - Forms with live validation
  - Everything, which needs in-browser event handlers

Mix SSR with Client Components

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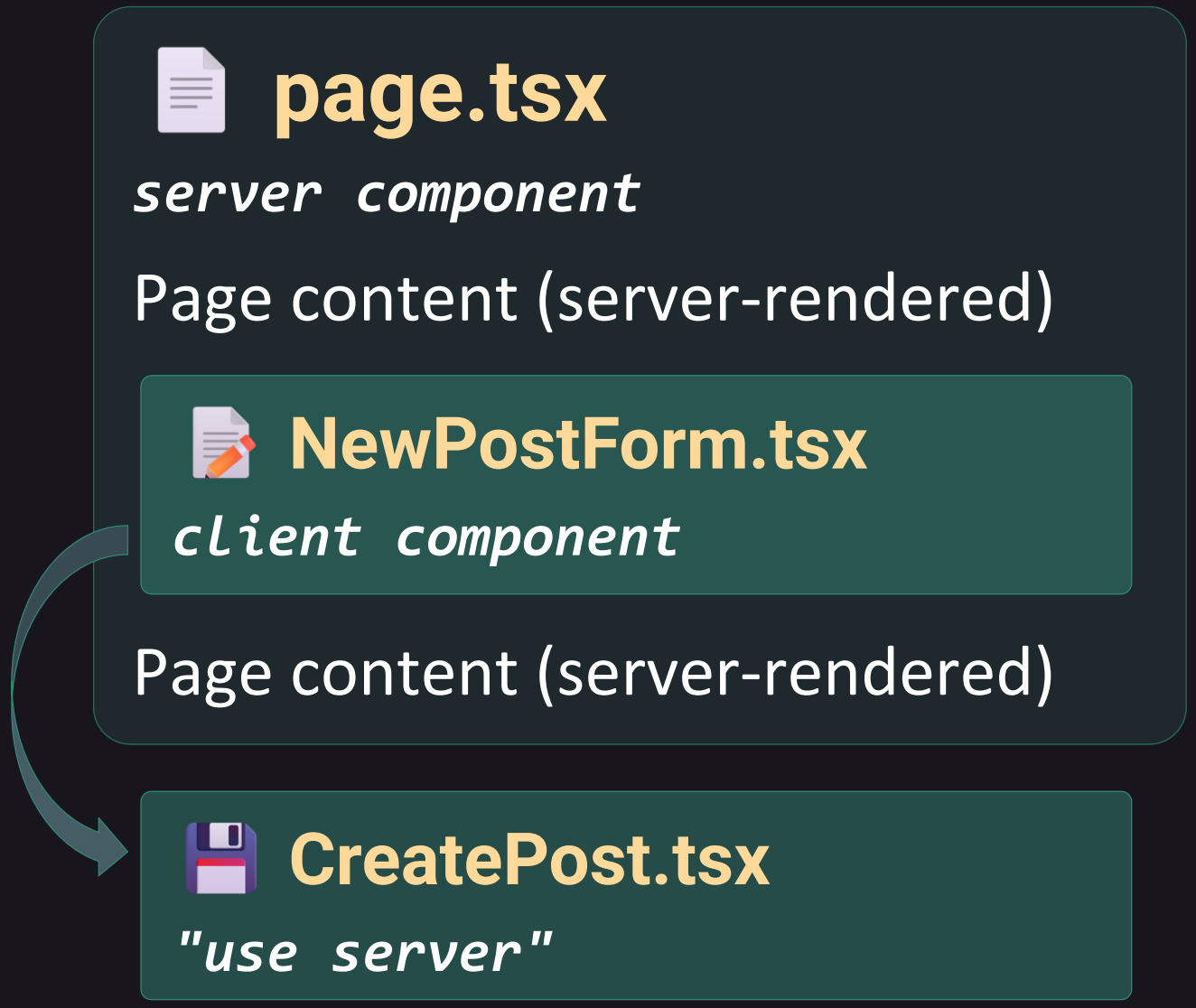
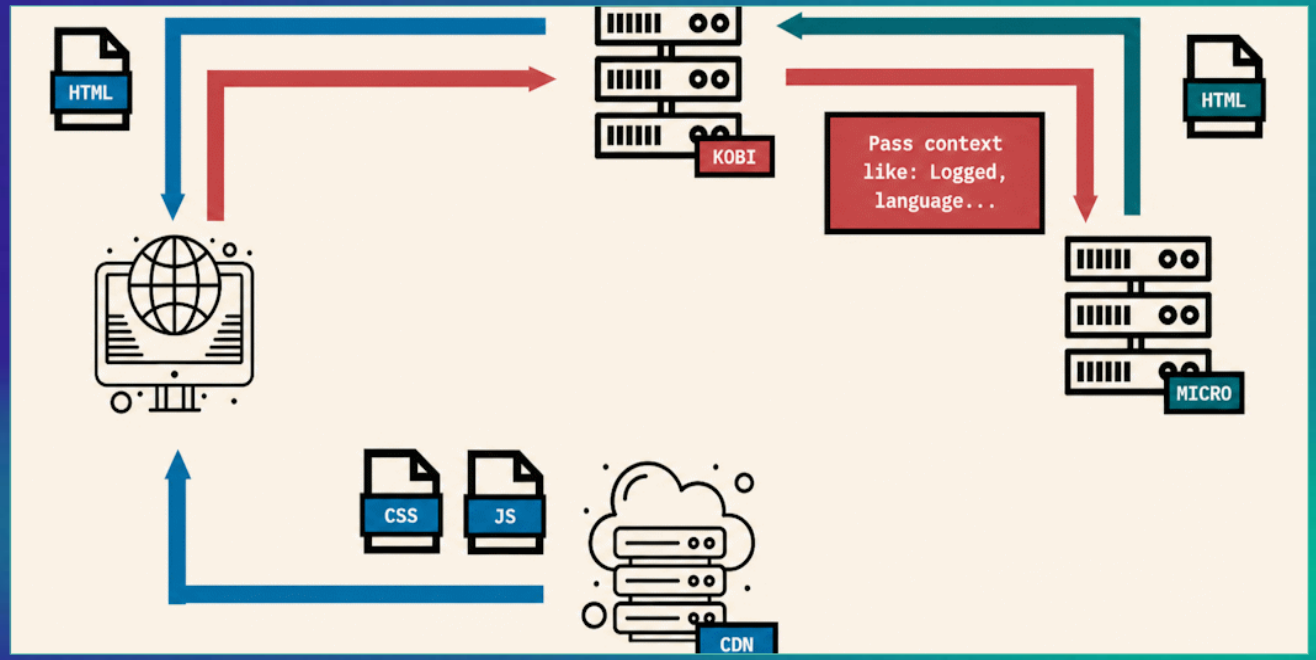
- How to implement a "Create a Post" page?
  - **page.tsx** → server component
  - **NewPostForm.tsx** → client component (interactive form)
  - **createPost.tsx** action → server-side mutation



Static, Dynamic and Hybrid Rendering

SSG, SSR, ISR, Caching and Revalidation



Three Server Rendering Strategies

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📄 SSG (Static Site Generation)

- HTML generated at **build time**, served from CDN
- **Fastest**, **cheapest**
- Perfect for **About**, **Docs**, **Landing Pages**

⚙️ SSR (Server-Side Rendering)

- HTML generated on **every request**
- Use for **user-specific** or **rapidly changing** data
- **Always fresh**, SEO-friendly

🔄 ISR (Incremental Static Regeneration)

- **Hybrid approach**: SSG + SSR combined
- Static, but **auto-revalidates** every X seconds
- Best of both: **fast AND fresh**

How Next.js Picks the Rendering Mode?

- Next.js decides **automatically** based on what the route uses:
  - Uses `cookies()`, `headers()`, or dynamic params → **SSR**
  - Uses `fetch(url, { next: { revalidate: 60 } })` → **ISR**
  - Everything static and cacheable → **SSG**
- Next.js rendering rules:
  - **Anything dynamic** in the route → whole route becomes dynamic (SSR)
  - **Everything cacheable** → route stays static (SSG)
  - Used `` `revalidate` `` → hybrid (ISR)

View Rendering Mode for Routes

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- See the **rendering mode** for each route in Next.js:

```
npm run build
```

```
PS nextjs-examples> npm run build

> nextjs-examples@0.1.0 build
> next build

▲ Next.js 16.2.4 (Turbopack)

  Creating an optimized production build ...
✓ Compiled successfully in 3.3s
✓ Finished TypeScript in 2.6s
✓ Collecting page data using 7 workers in 899ms
✓ Generating static pages using 7 workers (9/9) in 292ms
✓ Finalizing page optimization in 22ms

Route (app)                 Revalidate  Expire
┌ ○ /
├ ○ /_not-found
├ ○ /about
├ ƒ /api/products
├ ○ /client-comp
├ ○ /news                          5m      1y
├ ƒ /news/[id]
└ ○ /products

○  (Static)   prerendered as static content
ƒ  (Dynamic)  server-rendered on demand
```


```
Route (app)                 Revalidate  Expire
┌ ○ /
├ ○ /_not-found
├ ○ /about
├ ƒ /api/products
├ ○ /client-comp
├ ○ /news                          5m      1y
├ ƒ /news/[id]
└ ○ /products

○  (Static)   prerendered as static content
ƒ  (Dynamic)  server-rendered on demand
```

SSG – Static Page Example

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- **Static pages** (SSG)
  - **Built once**, served forever (until next deploy)
  - Cached by the CDN
  - Perfect for **rarely-changing** content
- Run `npm run build` → see `/about (Static)`
- **Default mode** if no dynamic APIs are used

```
// app/about/page.tsx → SSG by default
export default function AboutPage() {
  return (
    <main>
      <h1>About Us</h1>
      <p>Founded in 2013…</p>
    </main>
  );
}
```

SSR – Dynamic Page Example

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- **Dynamic pages** (SSR)
  - **Re-rendered** on every request
  - `cache: "no-store"` → always fresh
  - Each visitor sees **live data**
- `force-dynamic` makes a route SSR

```tsx
// app/now/page.tsx
export default async function NowPage() {
  const res = await fetch(
    "https://api.weather.com/Rome/now",
    { cache: "no-store" });
  const data = await res.json();
  return <h1>{data.temp} °C</h1>;
}
```

```tsx
// app/now/page.tsx
export const dynamic = "force-dynamic";
```

ISR – Hybrid Rendering

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- **Cached pages with revalidate** (ISR)
  - **Static page**, with periodic **refresh**
- First visit after **60s** triggers a rebuild
  - Everyone else gets the **cached version** → very fast
- **Best of both worlds**: fast AND fresh

```
// app/news/page.tsx
export default async function NewsPage() {
  const res = await fetch(
    "https://api.news.com/top",
    { next: { revalidate: 60 } }); // 60s
  const articles = await res.json();
  return (
    <ul>
      {articles.map(a =>
        <li key={a.id}>{a.title}</li>
      )}
    </ul>
  );
}
```

Live Demo – Compare SSG, SSR, ISR

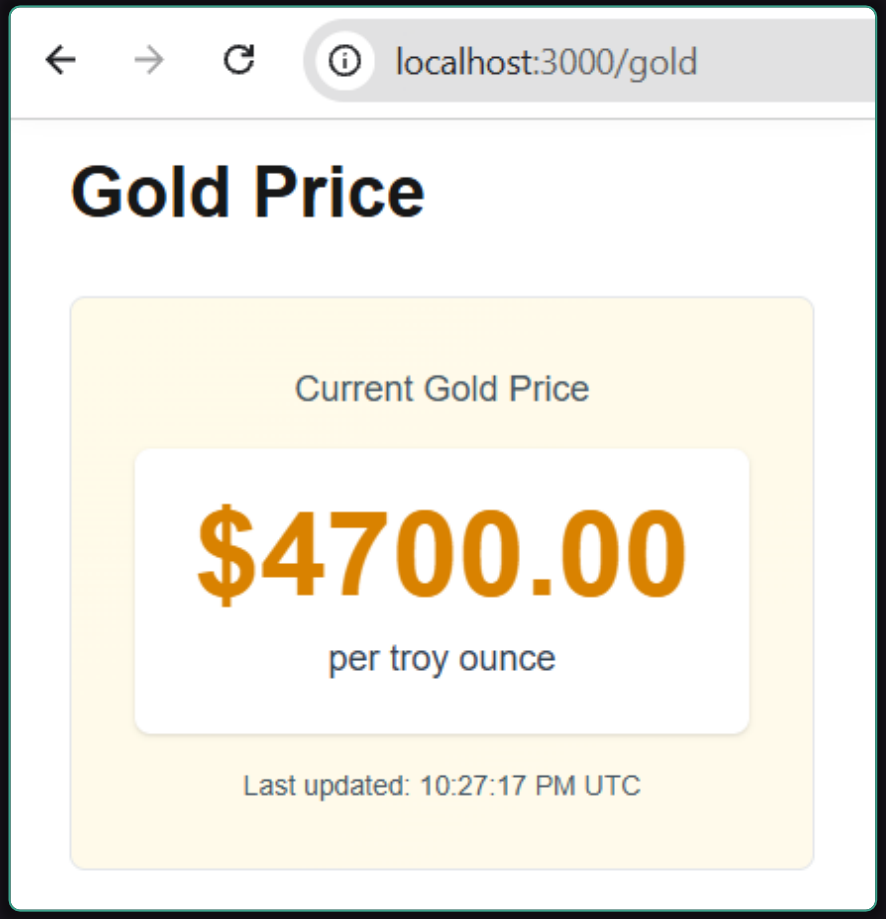
SoftUni AI

🤖 **AI prompt** to create static and dynamic pages:

Create **3 pages** in the same Next.js app:

- **/contacts** -> **static** (SSG), shows "**Contact Us**" (sample content)
- **/time** -> **dynamic** (SSR), shows the current **server time** on every refresh
- **/gold** -> **cached** (ISR), fetches **gold price** from https://api.gold-api.com/price/XAU, revalidates every 30s

🔍 Run `npm run build` to see **static / dynamic** routes



Server Actions in Next.js

Form Submissions without REST Endpoints

What are Server Actions?

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- ⚡ **Server actions** in Next.js
  - ⚡ Functions marked `"use server"` that run on the **server**
  - 📝 Called directly from **forms** or **client components**
  - 🚫 **No `/api` routes needed** for simple mutations
- 🛠️ **Server actions** can:
  - Access the **database** directly
  - Read **secrets and `.env` variables**
  - **Revalidate caches** after a mutation
  - **Redirect** after submission

```
"use server"
export async function
submitForm(form: FormData) {
  // db.insert (form data)
}
```

Server Action – Example

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- **Action file**: declare with `"use server"`
- **Page**: pass it to `<form action={…}>`

```
// app/dashboard/actions.ts
"use server";
import { db } from "@/lib/db";

export async function addBookmark(
    form: FormData) {
  const url =
    form.get("url") as string;
  await db.insert(bookmarksTable)
    .values({ url });
}
```

```
// app/dashboard/page.tsx
import { addBookmark } from
"./actions";

export default function
Dashboard() {
  return (
    <form action={addBookmark}>
      <input name="url" />
      <button type="submit">
       Add</button>
    </form>
  );
}
```

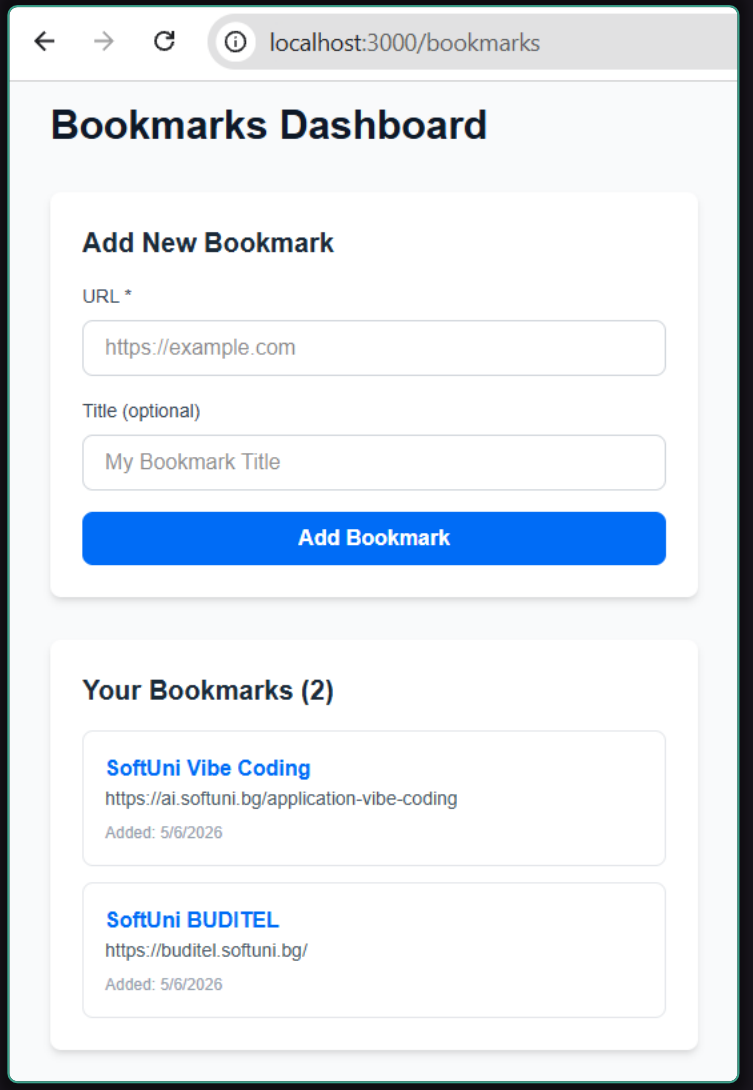
Live Demo – AddBookmark Form

SoftUni AI

🤖 **AI prompt** to build a dashboard with **server actions**:

Build a **bookmarks dashboard** in `app/bookmarks/page.tsx`:

- For simplicity, keep all bookmarks in a local storage file: `**app-data.json**`
- **Server component** lists all bookmarks from the storage
- `<form>` submits to **server action** `addBookmark`
- The action inserts the **URL into bookmarks**
- Call `revalidatePath("/dashboard")`
- No `useState()`, `useEffect()`, or `/api` route



Styling Next.js Apps with Tailwind

Utility-First CSS, Responsive Design, Component Styling

Tailwind CSS in Next.js

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- Tailwind = **utility-first CSS framework**
- **Pre-configured** by **`create-next-app`**
- Write styles as classes, **directly in JSX**
- **No separate CSS**, no naming conventions
- **Responsive** with `sm:`, `md:`, `lg:` prefixes

- **Tailwind** is quite suitable for **AI-first development**
  - Less files to manage → less complexity for the AI dev agents

```
<button className="
  bg-blue-600 hover:bg-blue-700
  text-white font-semibold
  px-4 py-2 rounded-lg
  transition-colors">
  Sign Up
</button>
```

Common Tailwind Utilities

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📐 Layout

```
flex
grid
gap-4
items-center
```

📏 Spacing

```
p-4
px-6
my-2
space-y-4
```

🔤 Typography

```
text-lg
font-bold
text-gray-700
leading-relaxed
```

🎨 Colors

```
bg-blue-500
text-white
border-gray-300
```

✨ Effects

```
shadow-md
rounded-xl
hover:scale-105
transition
```

📱 Responsive

```
text-base
md:text-lg
lg:text-xl
```

🌙 Dark mode

```
dark:bg-gray-900
dark:text-white
```

💡 Tip

Compose tiny utilities into rich UIs without custom CSS

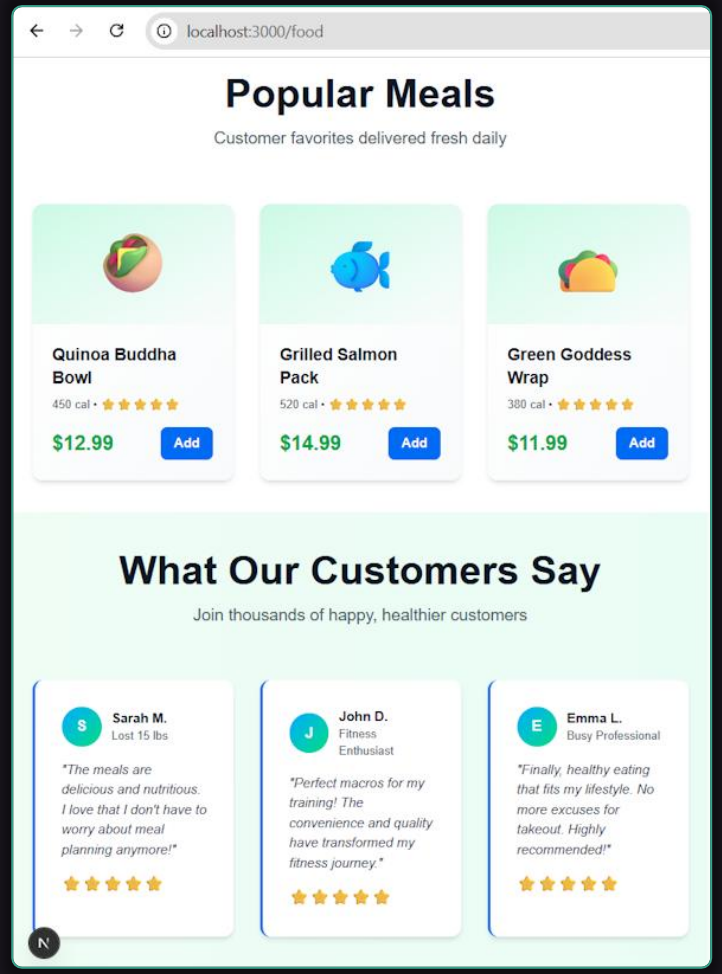
Live Demo – Landing Page with Tailwind

SoftUni AI

🤖 **AI prompt** to build a landing page with Tailwind:

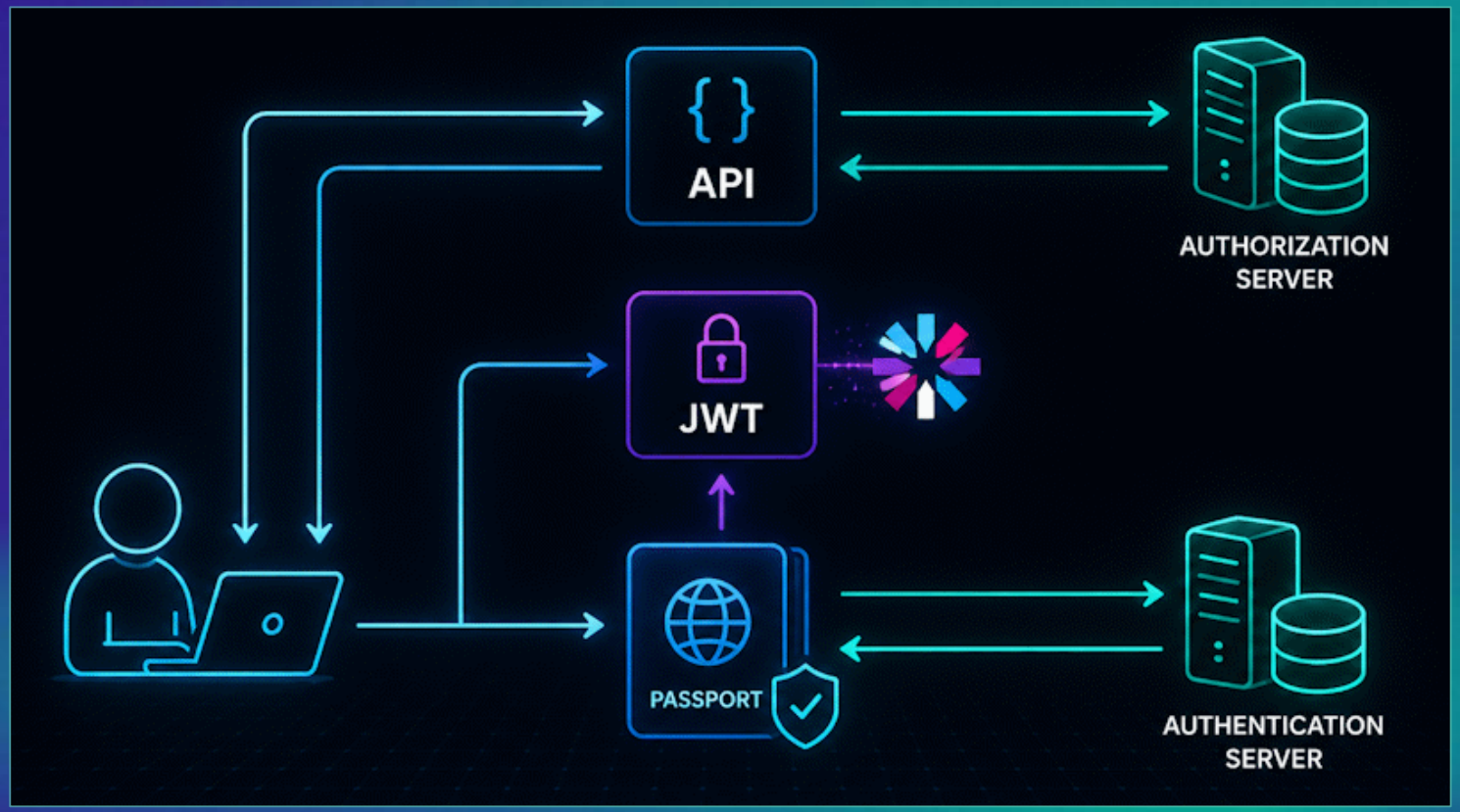
Create a modern, responsive **landing page** about "Healthy Food" business at `app/food/page.tsx`:

- Promote a healthy food and nutritious meals
- **Hero section**: large heading, subheading, CTA
- Sections: "**Features**" (3-column responsive grid), "**How It Works**", "**Popular Meals**", "**Testimonials**", "**Benefits**", "**Final CTA**"
- Implement **modern UI**, responsive design, with transitions, effects, icons / emojis, etc.
- Use only **Tailwind classes**, no custom CSS



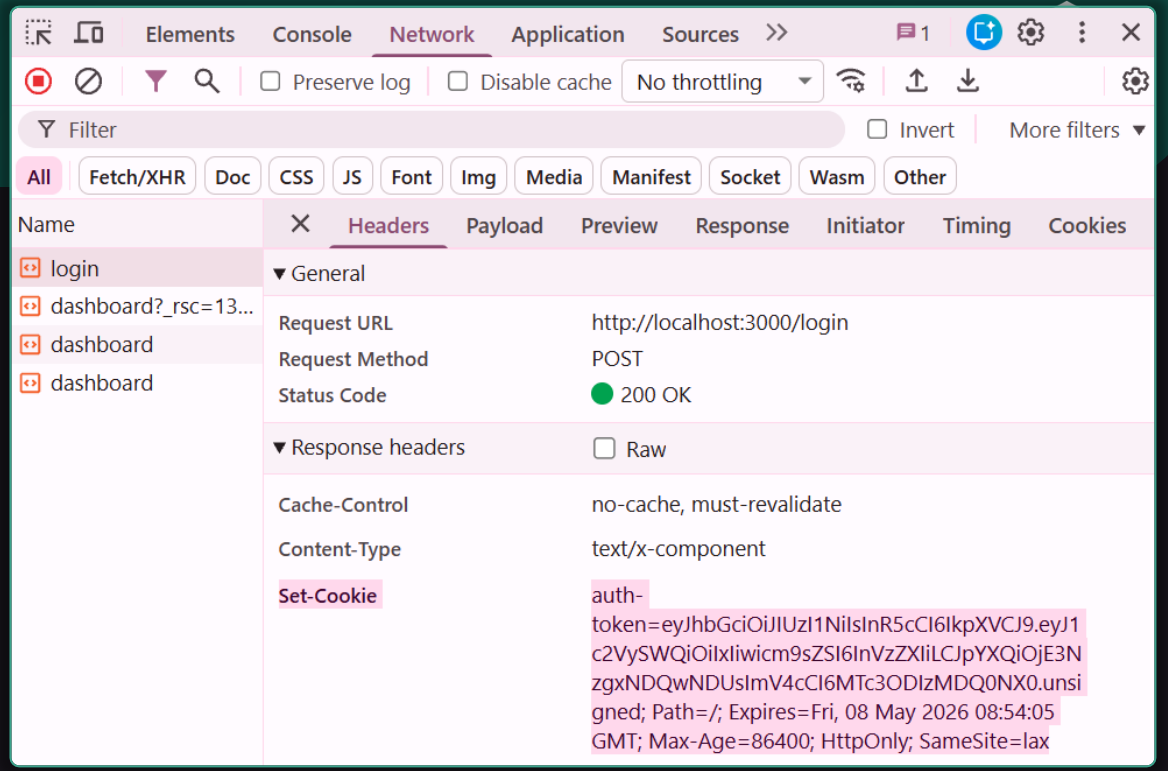
Auth, JWT and Middleware

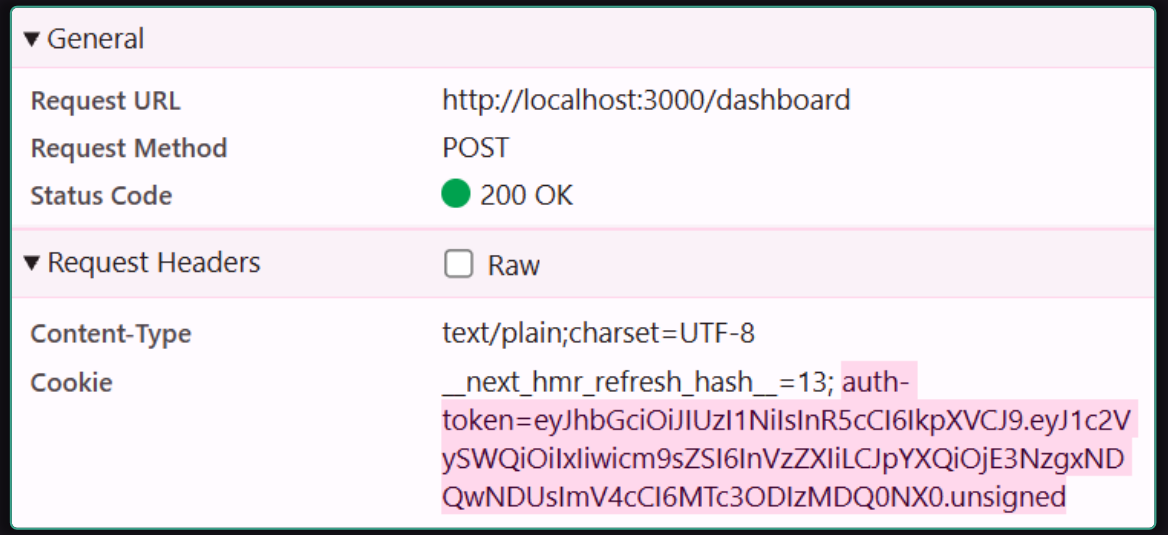
Cookies, Sessions, JWT, Role-Based Access



HTTP Cookies

- What are **HTTP cookies**?
  - Small **key-value data** strings
  - Stored in the **browser**
  - Sent by the **server** via a `**Set-Cookie**` HTTP response header
- **Request flow** with cookies
  - Browser **automatically attaches matching cookies**
  - To every subsequent HTTP request via the `**Cookie**` header





HTTP Cookies: Uses and Attributes

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- **Common uses** of cookies
  - Sessions, preferences, carts, tracking
- Cookie **key attributes**
  - **Expires** / **Max-Age**, **HttpOnly**, **Secure**, **SameSite**
- Cookie **scope**
  - **Domain** + **Path**
  - Restrict which origins receive them
- Learn more at: https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Cookies

Auth Concepts: Headers & Cookies

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- In software systems **auth** is typically done with **JWT tokens**

⚙️ RESTful APIs: **HTTP header** brings the auth token

HTTP Response (Login)
`POST /api/login { user, pass }`
`200 OK { "token": "…" }`

HTTP Request (with Auth)
`Authorization: Bearer <token>`
`GET /api/protected-resource`

🌐 Browser-based apps: **HTTP cookie** brings the auth token

HTTP Response (Login)
`POST /login { user, pass }`
`Set-Cookie: session=<token> …`

HTTP Request (with Auth)
`Cookie: session=<token> …`
`GET /dashboard/protected-page`

JWT Authentication Flow for Web

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1. Login
Email + password

2. Sign JWT
{ id + role } baked in

3. Set Cookie
httpOnly cookie

4. Auto-Send
Cookie on every request

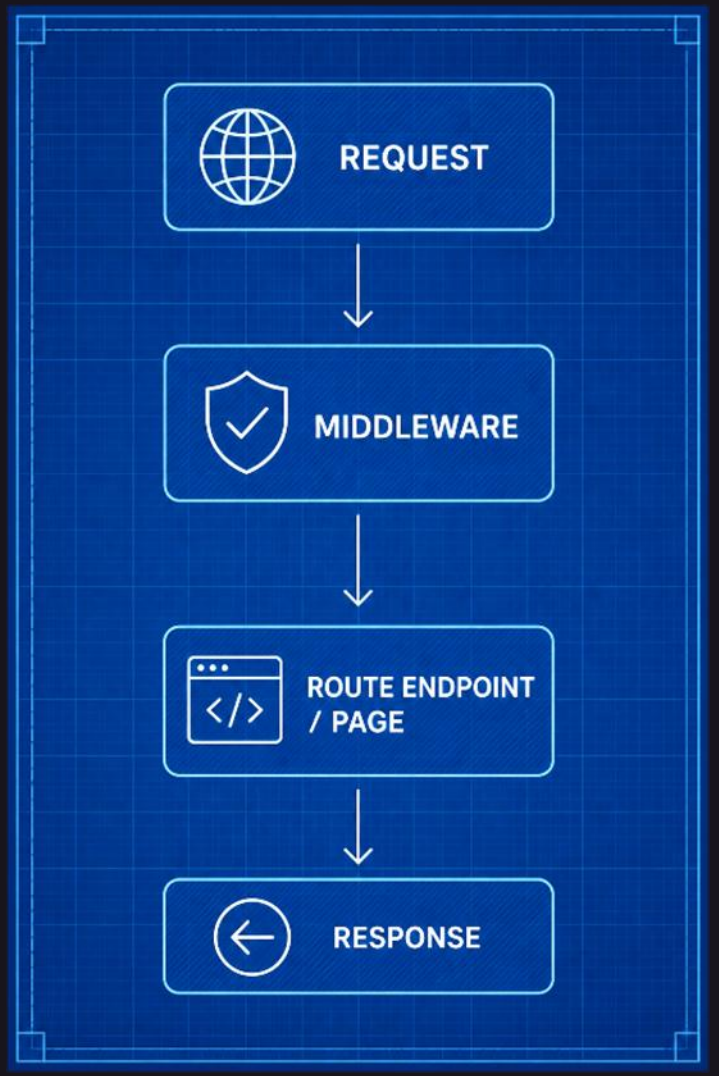
5. Decode JWT
Middleware reads it

6. Allow / Block
Based on role

7. Render UI
User-specific UI

Next.js Middleware

- In Next.js `middleware.ts` / `proxy.ts` runs **before every request** reaches a route
  - Placed at the **project root**
- Middleware **use cases**:
  - 🔐 **Authentication** (check JWT, redirect to login)
  - 👮 **Authorization** (role-based access)
  - 🧪 **A/B testing** (rewrite URLs)
  - 📊 **Logging**, analytics, geo-location
  - 🌍 **Locale detection** and redirects



Middleware – Example

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- Auth middleware
  - Reads `token` from **cookies**
  - Redirects unauthenticated users to **/login**
  - `matcher` limits which paths to process

```
// middleware.ts
import { NextResponse }
from "next/server";
import type { NextRequest
} from "next/server";
```

```
export function middleware(req: NextRequest) {
  let token = req.cookies.get("token")?.value;
  if (req.nextUrl.pathname.startsWith(
      "/dashboard")) {
    if (!token) {
      return NextResponse.redirect(
        new URL("/login", req.url));
    }
  }
  return NextResponse.next();
}

export const config = {
  matcher: ["/dashboard/:path*",
    "/admin/:path*"],
};
```

Reading User in Server Components

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- Server components can read `cookies()` directly
- Use `verifyJWT()` to decode the token
- **Note**: using `cookies()` makes the route **dynamic (SSR)**

```
import { cookies } from "next/headers";
import { verifyJWT } from "@/lib/auth";
```

```
export default async function Dashboard() {
  const token = (await cookies())
    .get("token")?.value;
  const user = token
    ? verifyJWT(token)
    : null;

  return (
    <h1>Hello, {user?.name ?? "Guest"}!</h1>
  );
}
```

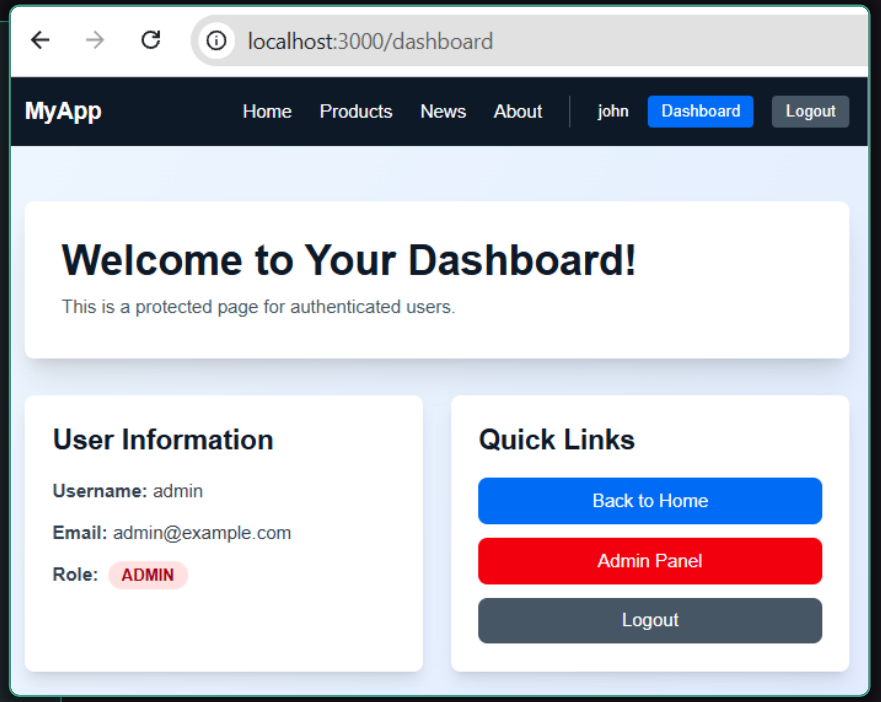
Live Demo – Role-Based App

SoftUni AI

🤖 **AI prompt** to implement users + roles with middleware:

In the **Next.js app** implement 3 areas:
- `/` → public home page
- `/dashboard` → protected, **logged-in** users
- `/admin` → protected, **admins** only

Auth setup:
- Use **server actions** for auth pages
- Keep users + roles in a local `users.json`
- `/login` sets a **JWT cookie** on success
- `middleware.ts` parses JWT → {`userId, role`}
- Redirects unauthenticated → `/login`
- Blocks non-admins from `/admin (403)`



Security Warning: JWT_SECRET

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- Critical **security warning**:
  - Who knows your **JWT_SECRET** can **login without a password**!
  - Initialize properly **JWT_SECRET** in **`.env`**
  - Make the app **fail** at runtime if **JWT_SECRET is missing**

TS auth.ts U

lib > TS auth.ts > str2ab

```
1   // Bad code --> JWT_SECRET should not be hardcoded
2   const JWT_SECRET = process.env.JWT_SECRET || 'your-secret-key-change-in-production';
```

Fix security: the app should fail to start if JWT_SECRET is missing in `.env`

Claude Haiku 4.5

Performance Optimizations

Link Prefetch, Image Optimization, Caching

Next.js is Built for Performance

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- Next.js is built for **performance** and **SEO**
  - **Server-side rendering** (SSR) → **loads faster** than client-side React
  - **Static pages** (SSG) and **cached pages** (ISR) → **served from CDN**
  - **Pre-fetched links** for faster navigation
  - **Image optimization** (smaller file size)
  - **Data caching** → reduce API / DB calls
  - **Suspense** and **streaming** → display the page in chunks (ready parts immediately, slower parts when loaded) → smoother UX

Faster Navigation with <Link>

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- Use `next/link` instead of `<a>` for internal navigation
- **Benefits** of **<Link>**
  - **Client-side page transitions** (no full reload)
  - **Automatic prefetching** when link enters the viewport
  - **Smooth**, **SPA-like UX**
- Only use `<a>` for **external links**

```
import Link from "next/link";

<Link href="/about">About Us</Link>

<Link
  href="/dashboard"
  prefetch={true}
>
  Dashboard
</Link>
```

Live Demo – Links in Next.js

SoftUni AI

- At the app Home page:
  - Insert **<Link href=…>**

```
<Link href="/about">About Us</Link>
```

  - Insert **<a href=…>**

```
<a href="/about">Slow Link: About</a>
```

  - Insert **<Link href=… prefetch={true}>**

```
<Link href="/contacts"
prefetch={true}>Contacts</Link>
```

- Note: **prefetch** works on **production** only

```
npm run build
npm run start
```

Image Optimization with <Image>

SoftUni AI

- `next/image` automatically optimizes images:
  - 🗜️ Converts to modern formats (**WebP, AVIF**)
  - 🧮 **Resizes** to device size (responsive)
  - 💤 **Lazy-loads** off-screen images
  - 📐 Prevents **layout shift** (requires width + height)

```tsx
import Image from "next/image";

<Image
  src="/hero.jpg"
  alt="Hero banner"
  width={1200}
  height={600}
  priority   // load immediately
/>
```

Caching in Next.js

SoftUni AI

- **Three cache layers**:
  - **Full Route Cache** – static HTML of SSG/ISR pages
  - **Data Cache** – results of **fetch()** calls (opt-in/opt-out)
  - **Request Memoization** – dedupe identical fetches
- **Revalidate on demand**:
  - `revalidatePath("/books")`
  - `revalidateTag("books")`

```
// Cache control on fetch()
fetch(url)
  // cached forever (default)
```

```
fetch(url, { cache: "no-store" })
  // never cached → SSR
```

```
fetch(url,
  { next: { revalidate: 60 } })
  // ISR, 60 seconds
```

```
fetch(url,
  { next: { tags: ["books"] } })
  // tag-based revalidation
```

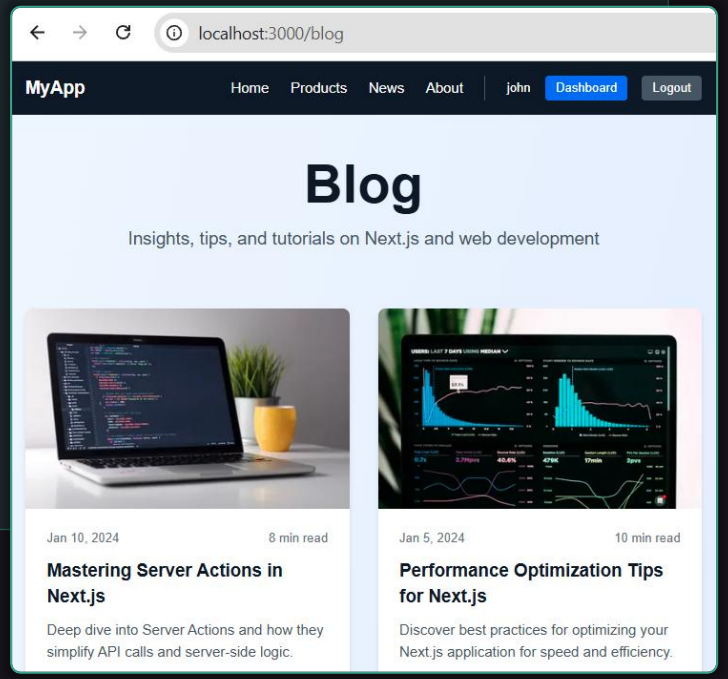
Live Demo – Optimized Blog Page

SoftUni AI

🤖 **AI prompt** to build a performance-optimized page:

Build a performance-optimized **blog index page**: **/blog** (SSR)

- Grid of **12 article cards**: image + title + excerpt
- `<Image>` for thumbnails (WebP, lazy loading)
- `<Link>` for each card -> **/blog/[slug]**
- **Prefetch** visible cards
- For simplicity, use `**blog-data.json**` (no database)



- **Deploy** the blog to Netlify
- Measure **page speed**: https://pagespeed.web.dev

Optimized Blog in Page Speed

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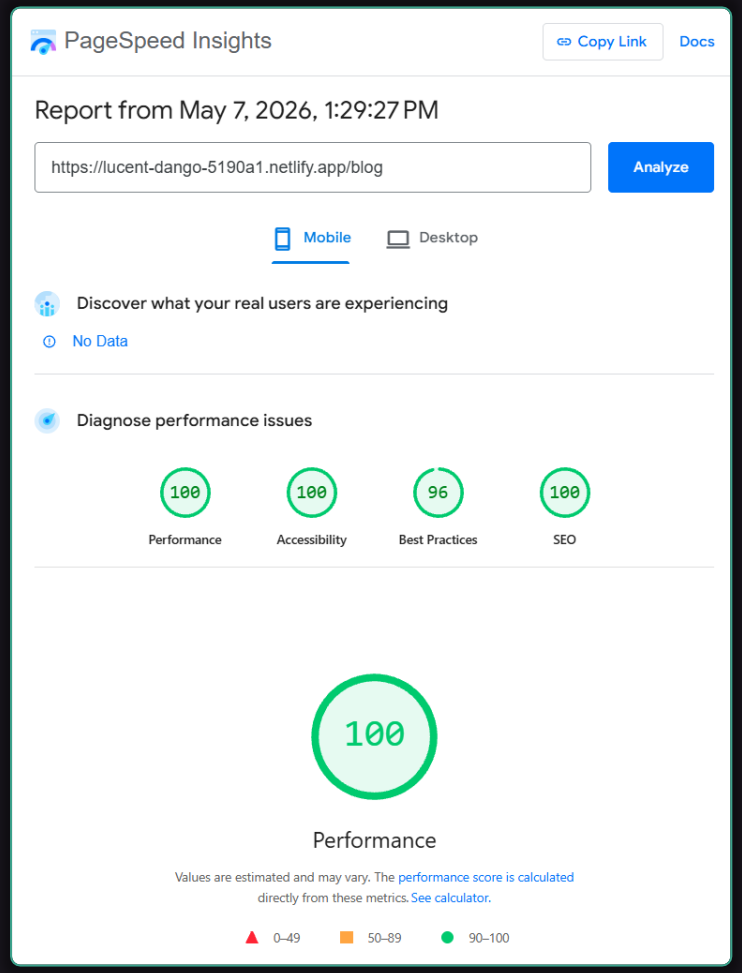
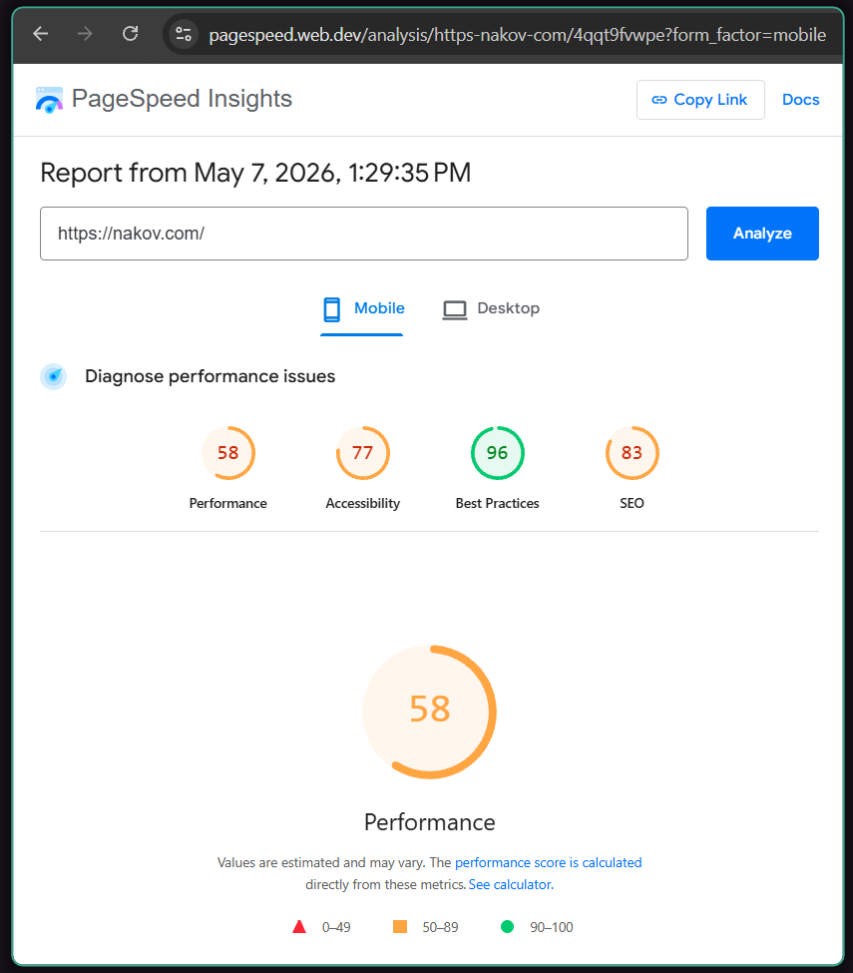
- Next.js site @ Netlify

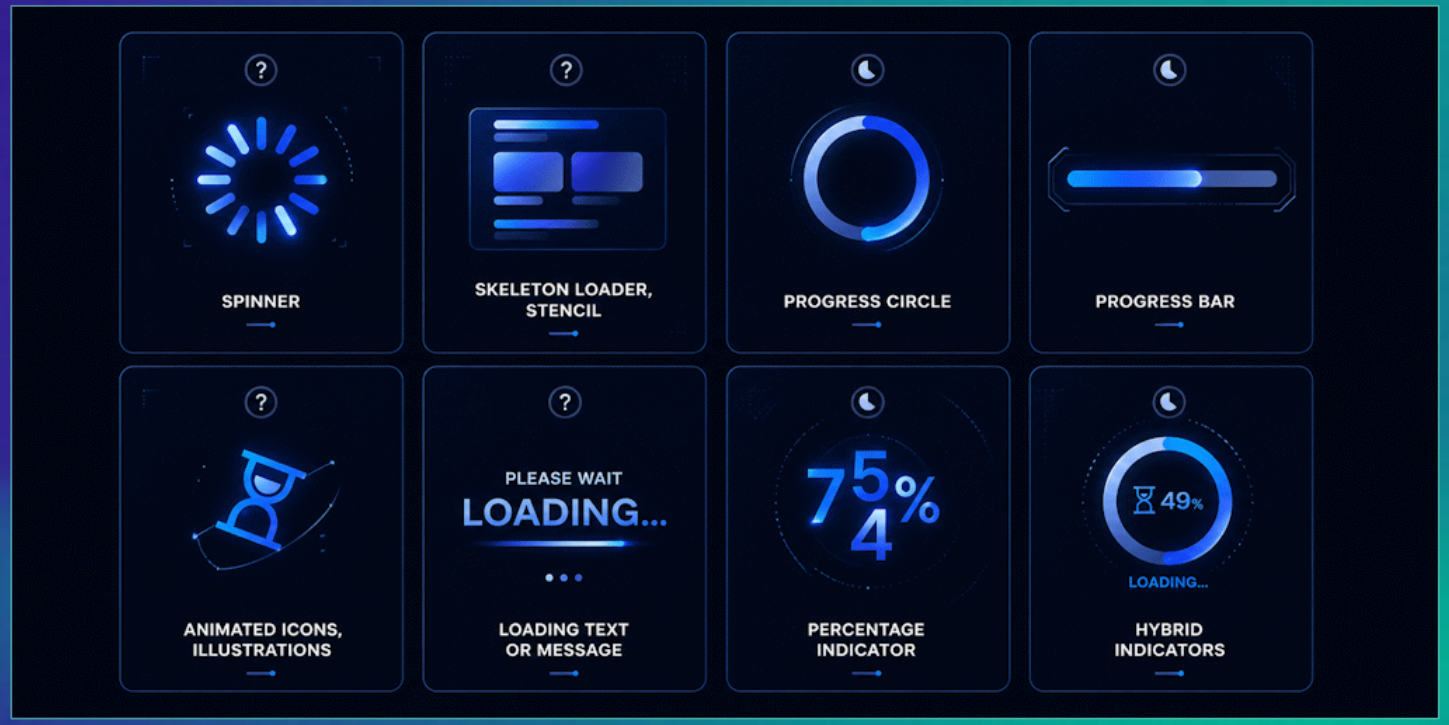


- WordPress site @ VPS



Suspense and Streaming

Progressive HTML, Loading States, Better UX



Why Suspense and Streaming?

SoftUni AI

⚠️ Problem

One **slow component** blocks the **whole page** from rendering

💡 Solution

Split the page into **chunks** and **stream** them as they become ready

⏸️ Suspense

Decides what can **wait** and what to **show** in the meantime

📡 Streaming

Decides when HTML **chunks are sent** to the browser

🚀 Result

User sees **fast content immediately**

Slow parts fill in **progressively**

<Suspense> in Next.js

SoftUni AI

- **Wrap slow components with <Suspense>** to stream loading
- Example:
  - Header shows **instantly**
  - Chart **streams in** when ready
- Each **<Suspense>** = independent stream

```
import { Suspense } from "react";
import SlowChart from "./SlowChart";
import FastHeader from "./FastHeader";
```

```
export default function DashboardPage() {
  return (
    <>
      <FastHeader />
      <Suspense
fallback={<p>Loading…</p>}>
        <SlowChart />
      </Suspense>
    </>
  );
}
```

Loading UI with `loading.tsx`

SoftUni AI

- A special file `loading.tsx` auto-wraps the route in `<Suspense>`
- **Shown instantly** while `page.tsx` fetches data on the server
- **Zero JS** needed – works with **Server Components**

```tsx
// app/dashboard/loading.tsx

export default function Loading() {
  return (
    <div className="p-8 animate-pulse">
      <p>Loading dashboard…</p>
    </div>
  );
}
```

- **Route-level** loading **progress indicator** (for all route pages)

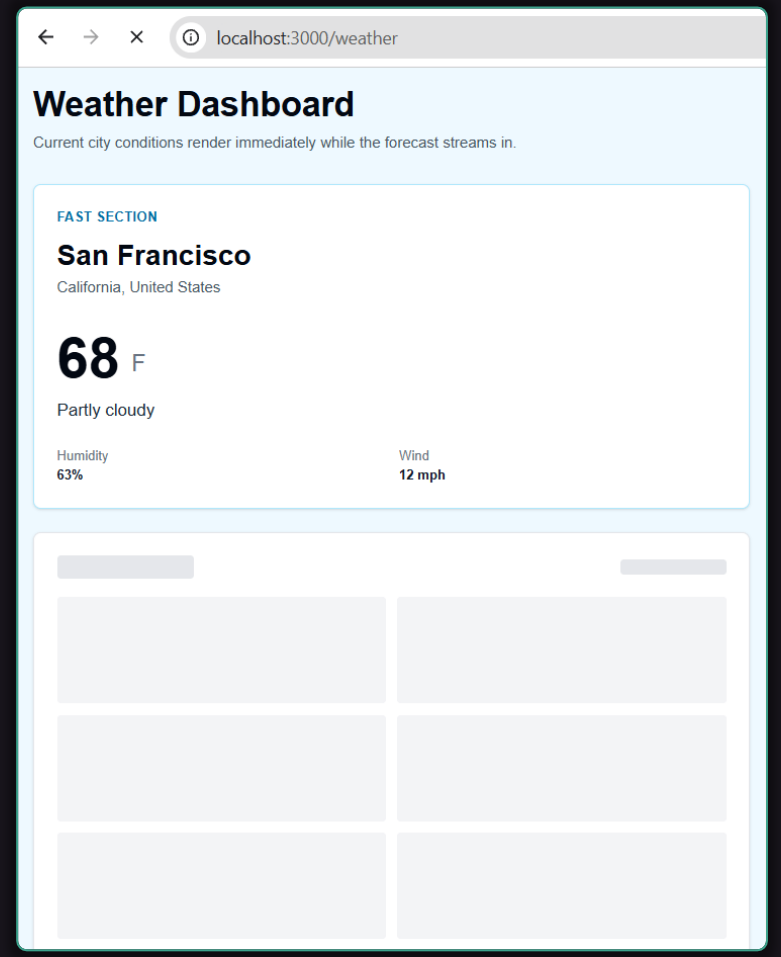
Live Demo – Weather Dashboard

SoftUni AI

🤖 AI prompt to build a dashboard with <Suspense>:

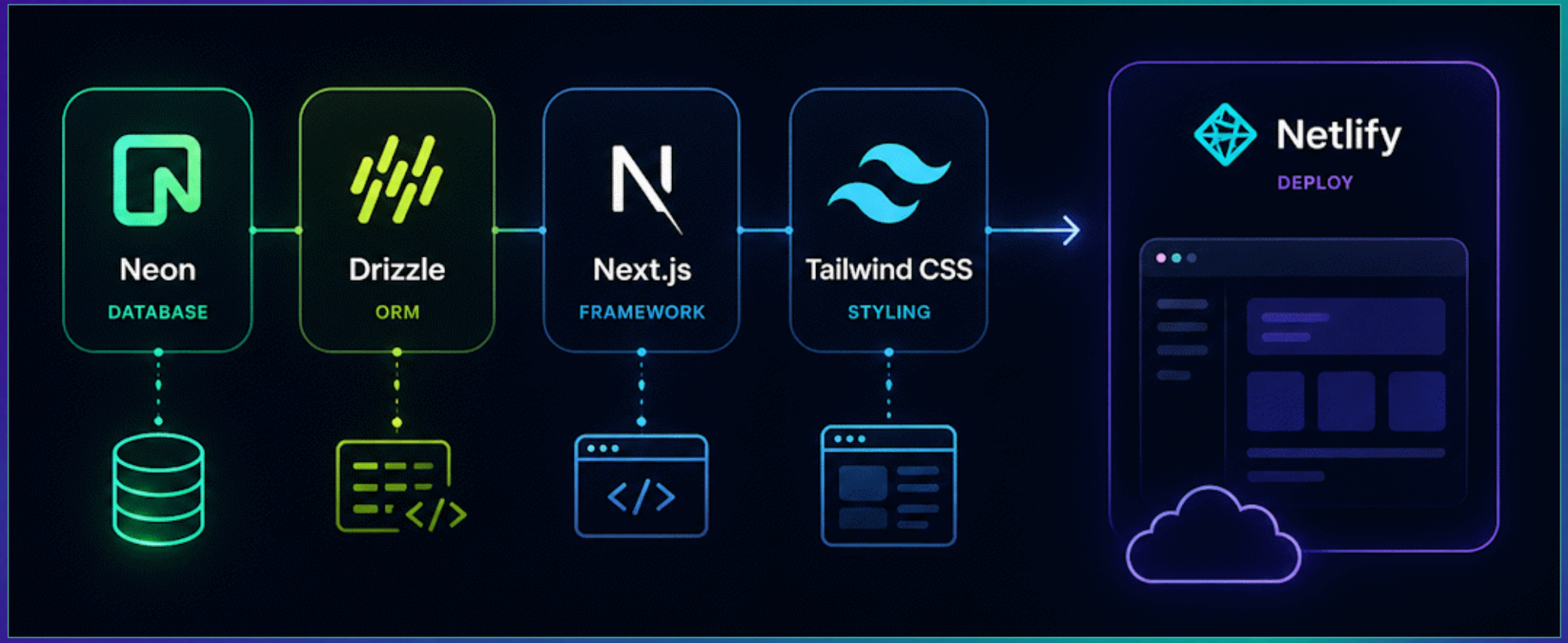
Build a **weather dashboard** at `/weather`:

- **Fast section**: city (renders instantly)
- **Slow section**: 7-day forecast (intentionally wait 3s at server-side)
- Wrap slow parts in `<Suspense>` with skeleton fallbacks
- Add `loading.tsx` as fallback
- For simplicity, use `**weather-data.json**` (no database)



Full-Stack App with Next.js

Next.js + Tailwind + Drizzle + Neon → Netlify



App Overview – "BlogHub" 

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- What shall we build?
- **Full-stack blog app**:
  - 🛠️ **Back-end**: TypeScript + Next.js (server components) + Drizzle ORM + Neon PostgreSQL
  - 🎨 **Front-end**: TypeScript + Next.js + React + Tailwind
- **Features**:
  - 📋 List all blog posts (**dynamic route, ISR**)
  - 📖 View a single post (**dynamic route, ISR**)
  - ✍️ Register, Login (**server action + form**)
  - 📝 Create a new post, 🗑️ Delete a post (**server action + form**)

Step 1: Create Next.js Project

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- Create a project in VS Code: `**nextjs-blog**`
- Scaffold an empty **Next.js project**:

```
npx create-next-app@latest .
```

```
PS nextjs-blog-example> npx create-next-app@latest .
>>
√ Would you like to use the recommended Next.js defaults? » Yes, use recommended defaults
Creating a new Next.js app in C:\Projects\Full-Stack-Apps\backend-apis\nextjs-blog-example.

Using npm.

Initializing project with template: app-tw
```

Step 2 – Create a Neon Database

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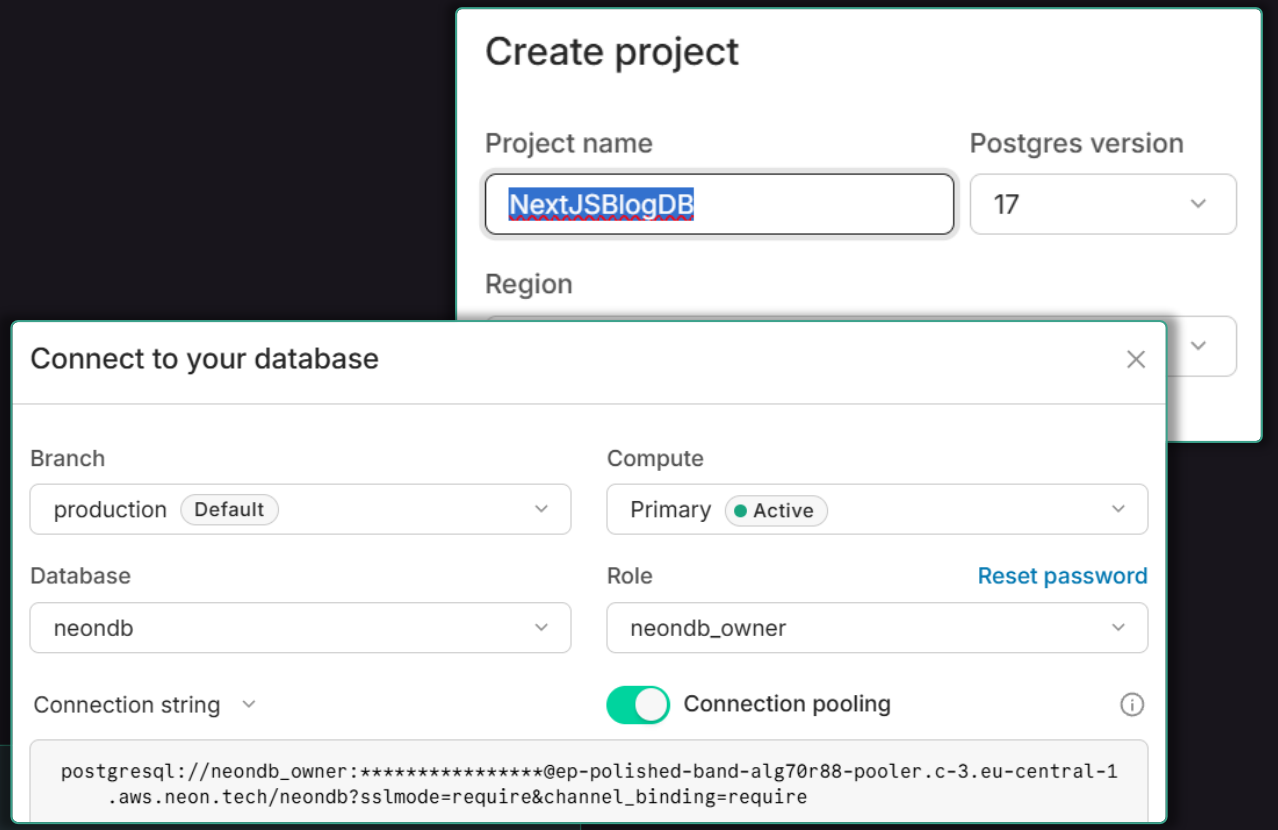
- Go to https://neon.tech
  → sign up (free tier)
- Create a project:
  NextJSBlogDB
- Copy the connection string → paste into `.env`



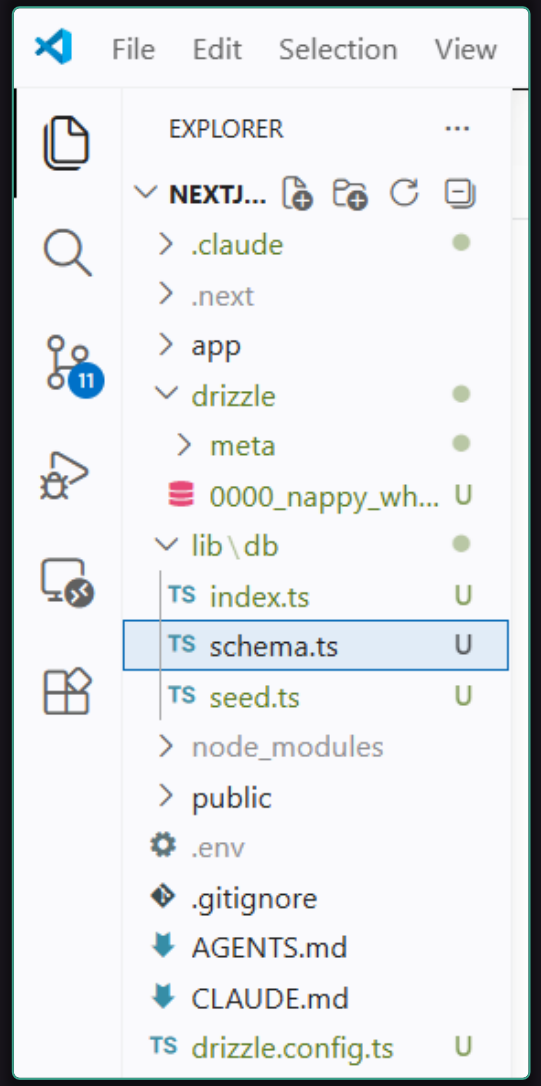

```
# .env
DATABASE_URL="postgresql://user:pass
  @ep-xxx.neon.tech/NextJSBlogDB"
```

Step 3 – Drizzle ORM Setup

SoftUni AI

🤖 **AI prompt** to setup Db with Drizzle:

Set up **Drizzle ORM** in this project:
- Install **Drizzle** ORM + **Neon** driver from NPM
- Create **Drizzle schema**:
  - **users** (email, passwordHash)
  - **posts** (title, content, tags[], date, owner → user)
- Create `**db:generate**` and `**db:migrate**` npm scripts (use Drizzle Kit)
- **Generate** an **migrate** the schema to DB



Step 4 – Seed Sample Data in DB

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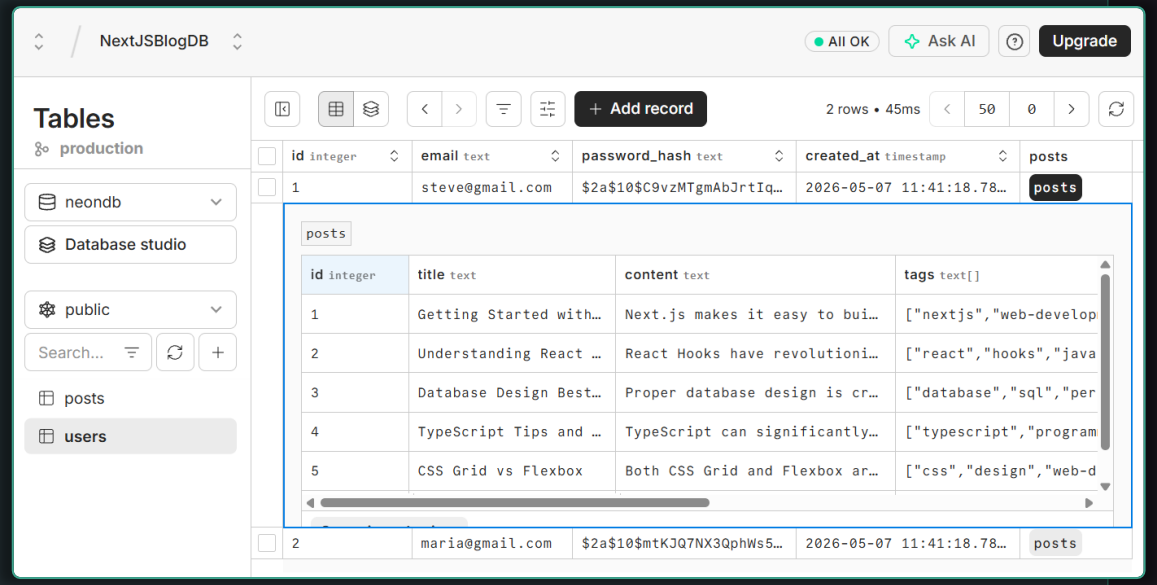
🤖 **AI prompt** to seed sample data in the DB:

Seed sample data in the DB:

- Define npm script `**db:seed**`
- Create **users**:
  { steve@gmail.com, pass123 },
  { maria@gmail.com, pass123 }
- Hash the passwords with **bcrypt**
- Insert 5-6 **blog posts** per user
- **Run** the DB seed script to seed DB



Step 5 – Build App Pages + Actions

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🤖 **AI prompt** to build the app functionality:

Build these **pages and routes**, using Next.js App Router (layout + pages + server actions):
App pages:
/ -> list all posts (ISR, 60s)
/posts/[id] -> single post (ISR, 60s)
/register, **/login**, **/logout** -> auth (SSG + client form + server action)
/posts/new -> form to create a post
Server actions:
- **login** / **register** / **logout** -> JWT token, bcrypt
- **createPost(formData)** -> insert, revalidate /, /posts
- **deletePost(id)** -> delete, revalidate /, /posts
Style with **Tailwind**: modern, responsive design, with **suspense** + **loading.tsx**

Step 5 – Deploy to Netlify 🚀

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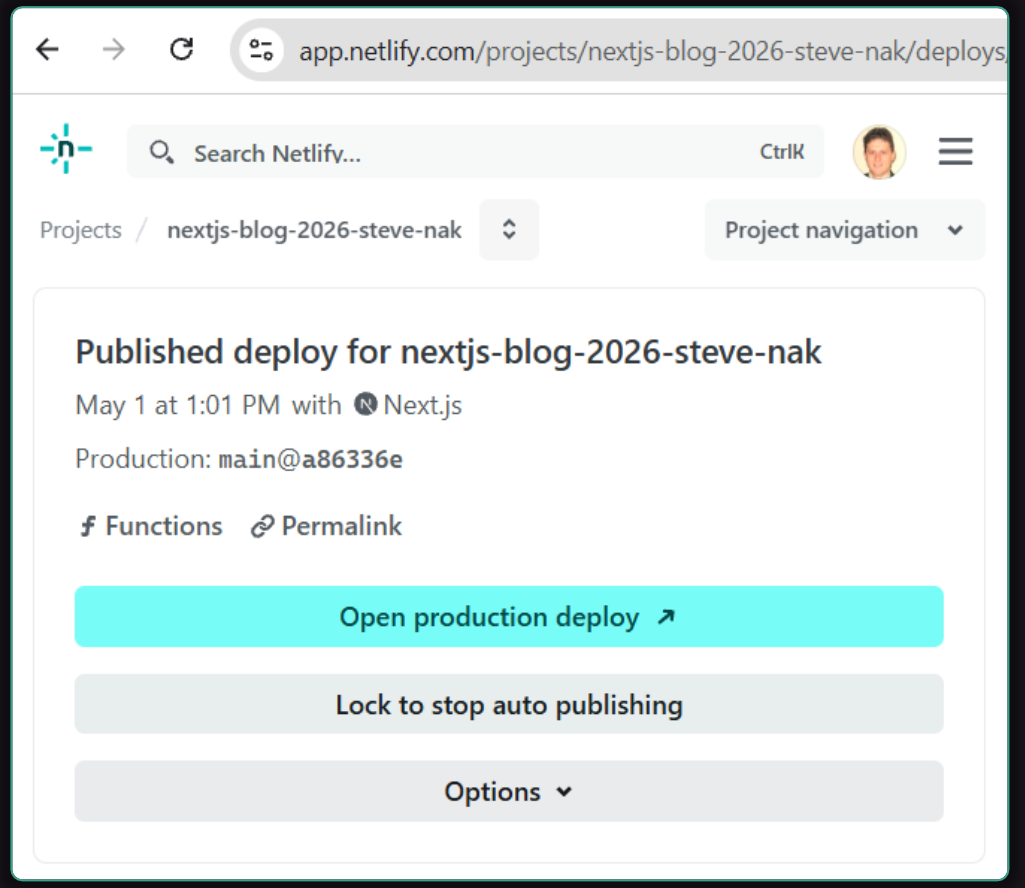
- Push the repo to **GitHub**
- Go to **https://netlify.com** → "Add new site" → "Import from Git"
- Netlify **auto-detects Next.js**
- **Add environment variables**:
  - `DATABASE_URL` = Neon connection string
  - `JWT_SECRET` = random secret
- Click **"Deploy"** → live at `<name>.netlify.app`
- Every `git push` triggers a re-deploy



Lesson Summary

SoftUni AI

- **Next.js** = full-stack React framework with **SSR**, **SSG**, **ISR**
- **App Router** uses file-system routing: folders = routes
- **Server Components** by default — DB access, SEO-friendly
- **Client Components** (`"use client"`) — interactivity, hooks, events
- **Server Actions** replace most `/api` routes for mutations
- **Middleware** runs before every request — perfect for **auth**
- `<Link>` and `<Image>` give free performance wins
- **Suspense + streaming** deliver fast UX even with slow data
- Deploy to **Netlify / Vercel** with zero config

Questions?

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